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Lecture 12

Optimization and Gradient Descent

The core algorithm of modern machine learning

EECS 189/289, Fall 2025 @ UC Berkeley
Joseph E. Gonzalez and Narges Norouzi

Key Factors in Generative AI Revolution

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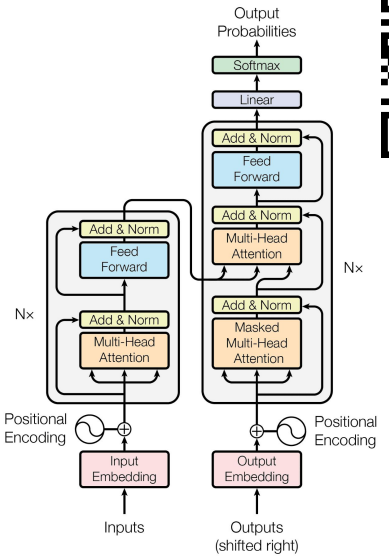
Data

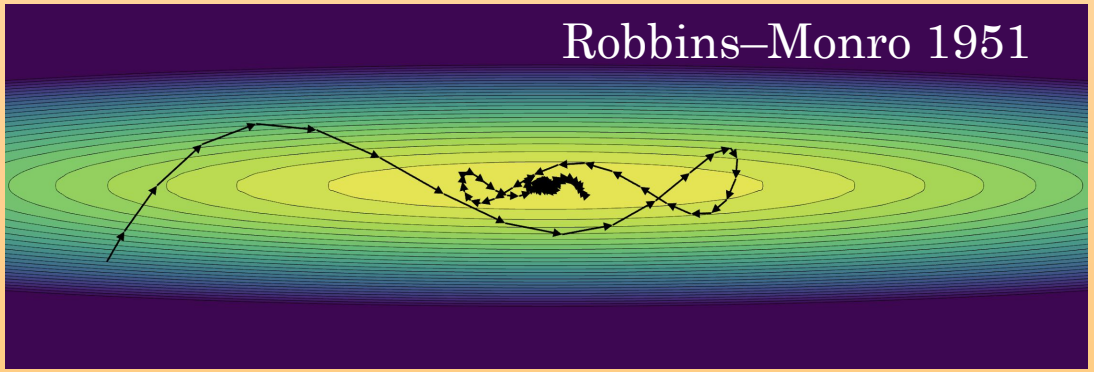
$$\ln p\left(\mathrm{Word}_n \mid \{\mathrm{Word}_i\}_{i=1}^{n-1}\right)$$

Loss Functions

Compute



Neural Architectures



Stochastic Gradient Descent

PyTorch

TensorFlow

Automatic Diff.

The General Optimization Problem

The General Optimization Problem

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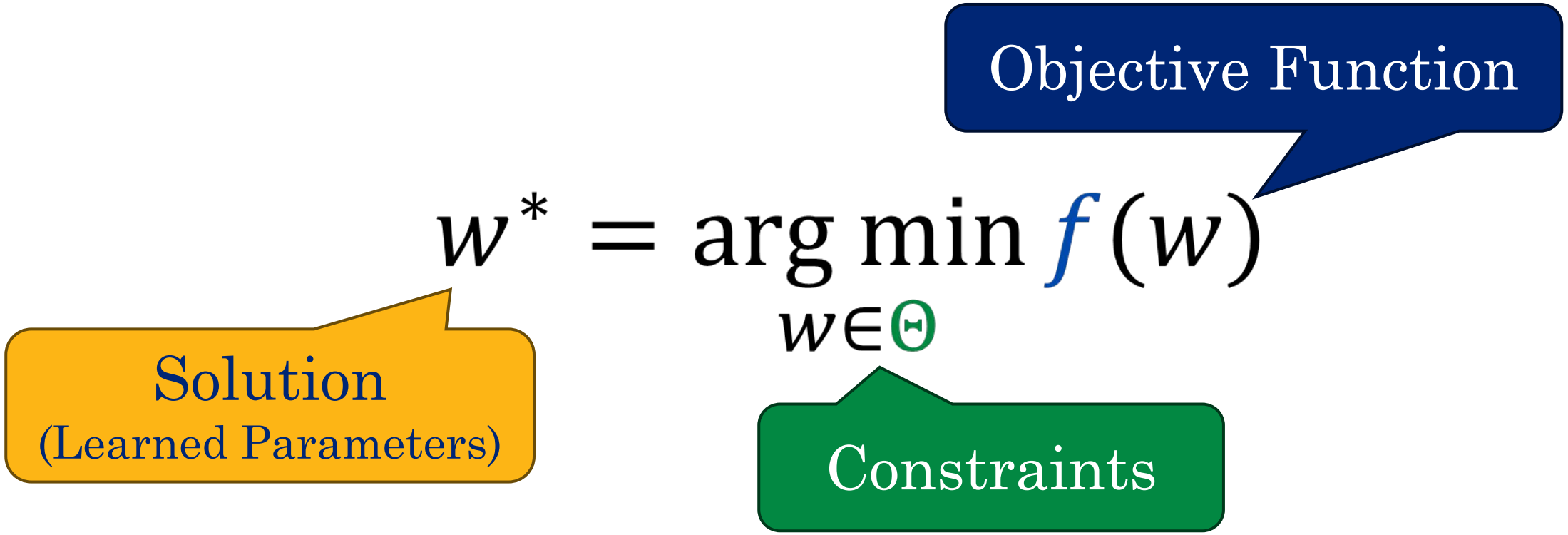
Most problems in ML can be reduced to the following general **optimization problem**



Optimization Constraints

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The **constraints** define the valid set of parameters.

$$w^* = \arg\min_{w \in \Theta} f(w)$$

Constraints

For the rest of this lecture, we will focus on the **unconstrained setting.**

$$\Theta = \mathbb{R}^d$$

The most common case in ML is **unconstrained optimization** in which $\Theta = \mathbb{R}^d$.

In some cases, there may be integer (e.g., $\Theta = \mathbb{Z}^d$) or probability constraints:

$$\Theta = \left\{ w \mid w \in \mathbb{R}^d, 0 \leq w \leq 1, \sum w_i = 1 \right\}$$

The Objective Function

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The **shape** of the objective function plays a big role in the complexity of solving the optimization problem.

Objective Function

$$w^* = \arg\min_{w \in \Theta} f(w)$$

The **Objective Function** in ML is often referred to as the **loss** or **error** function and depends on the **data.**

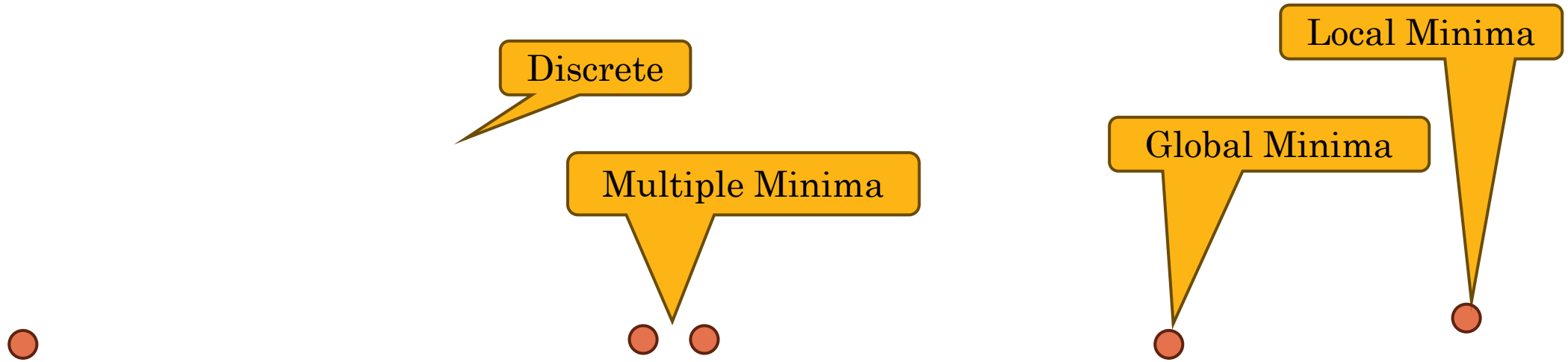
$$f(w) = E[w; \mathfrak{D}]$$

Examples Optimization Problems

$$\arg\min_{w \in \mathbb{R}} w^2 - 3w + 4$$

$$\arg\min_{w \in \mathbb{Z}} w^2 - 3w + 4$$

$$\arg\min_{w \in \mathbb{R}} w^4 - 5w^2 + w + 4$$



How do we solve these optimization problems?

What properties of the objective function make it "easier" to solve?

Plots in demo notebook.

Convex Functions

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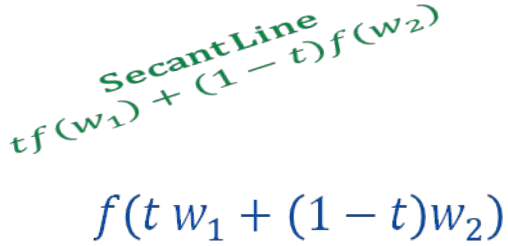
Definition: A function is **convex** on the domain Θ if for any pair of points $\forall\, w_1, w_2 \in \Theta$ and for all $0 \leq t \leq 1$:

$$f(t\, w_1 + (1-t)w_2) \leq tf(w_1) + (1-t)f(w_2)$$

Convex:



Non-Convex:

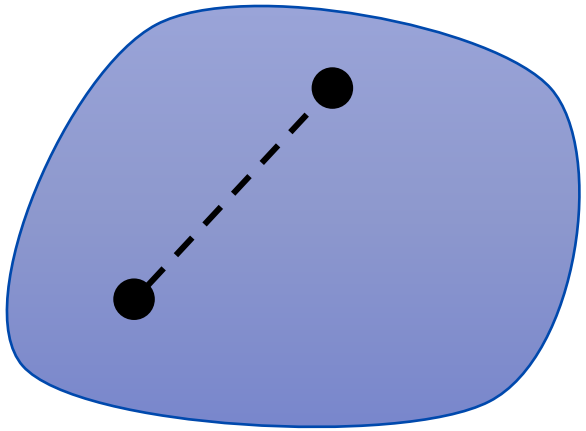
In general, there are **efficient algorithms** to reliably minimize convex functions.

Convex (Constraint) Sets

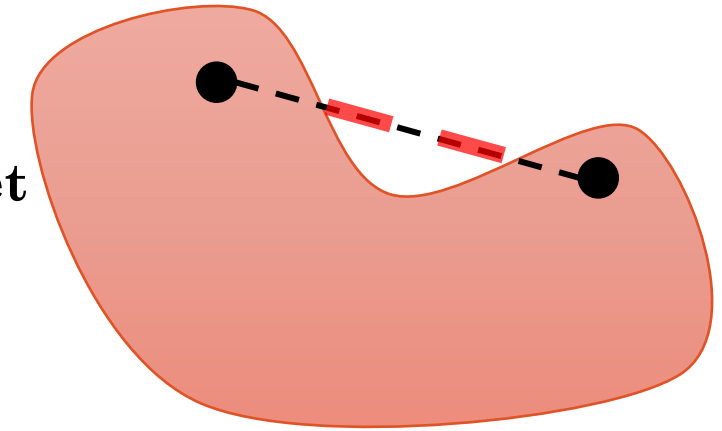
Definition: A set S is convex if it contains the **line segment** between **any two points** in the set:

$$\forall w_1, w_2 \in S, ;\ 0 \leq t \leq 1 :\quad t\, w_1 + (1-t) w_2 \in S$$

Convex Set



Non-convex Set



In general, there are **efficient algorithms** to reliably minimize **convex functions** over **convex sets**.

Optimization Problems

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Optimization of a **convex objective function** over a **convex constraint set**.

- *ML problems that we can solve efficiently*
- Primary focus of EECS-127 and books

Convex Function

Convex Constraints

Many classic ML problems are **convex**.

- **Least Squares Regression** and **Logistic Regression**

Most of **modern ML (deep learning) is not convex**.

- We use convex optimization **techniques** for non-convex problems
- In this class, introduce basic **optimization concepts**

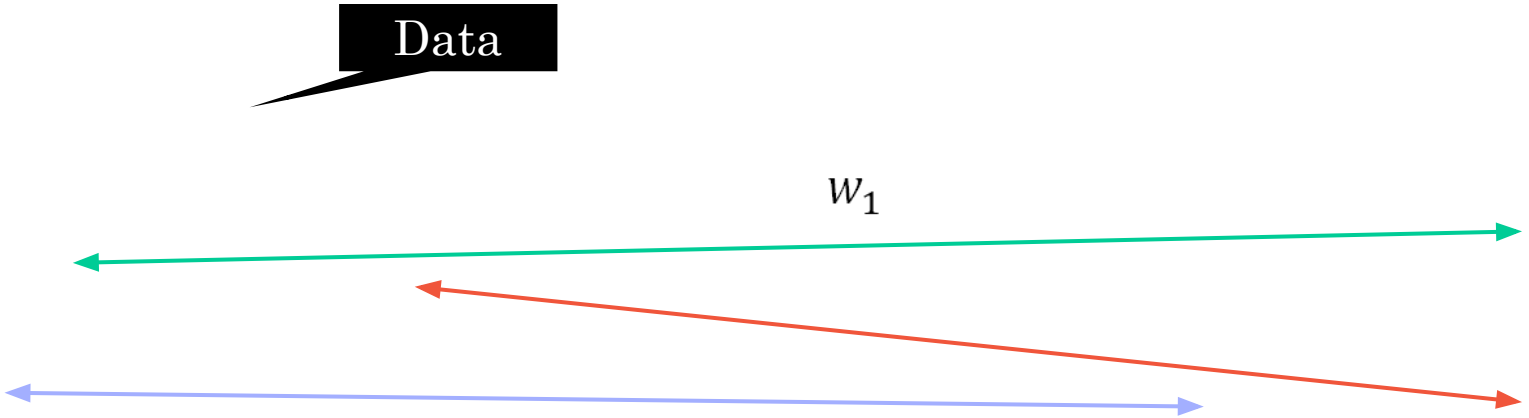
Minimizing the Error & Optimizing in Weight-Space

Points on the Error Surface

- Each **parameterization of a model** corresponds to a point on the error surface $E(w; D)$.

w_1

w_0

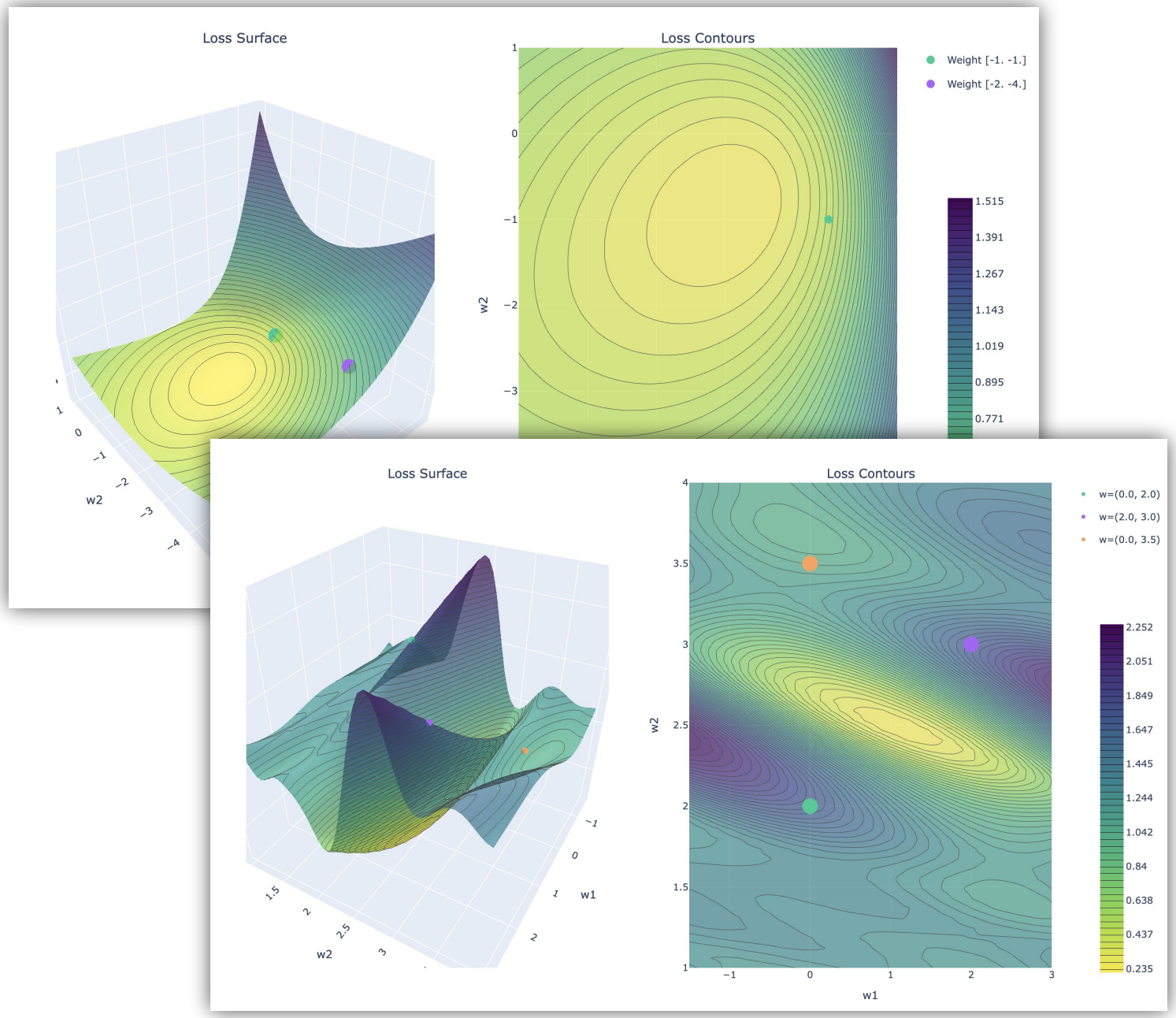


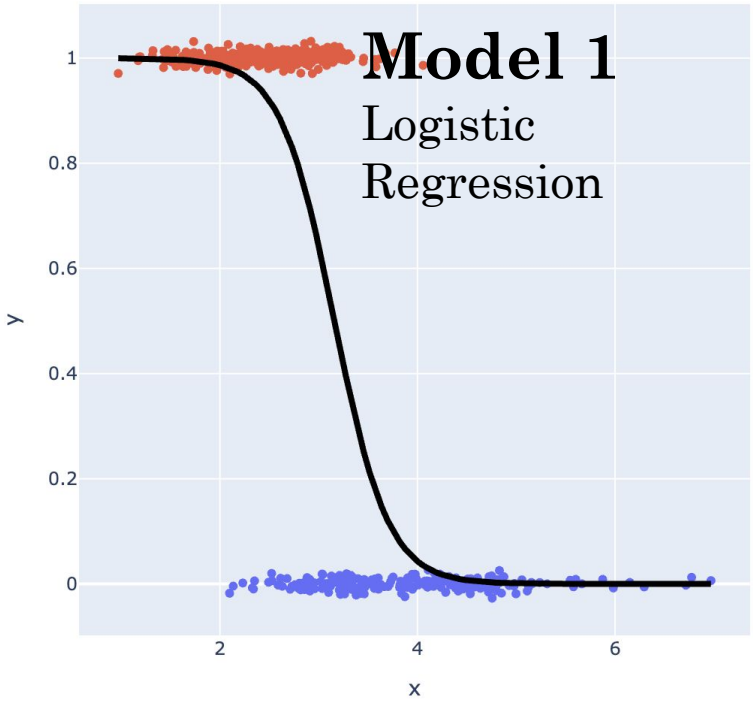
Goal: find parameterization with lowest error: $w^* = \arg\min_{w \in \mathbb{R}^d} E(w; D)$

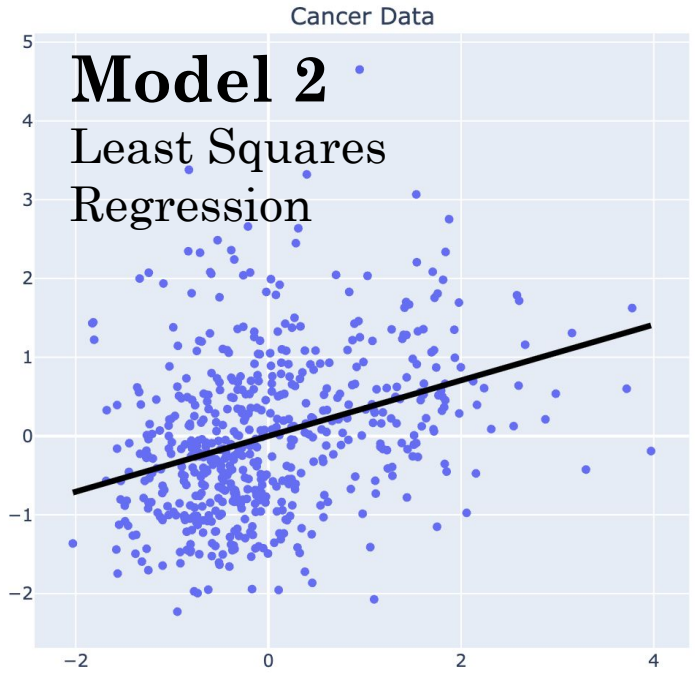
Plots in demo notebook.

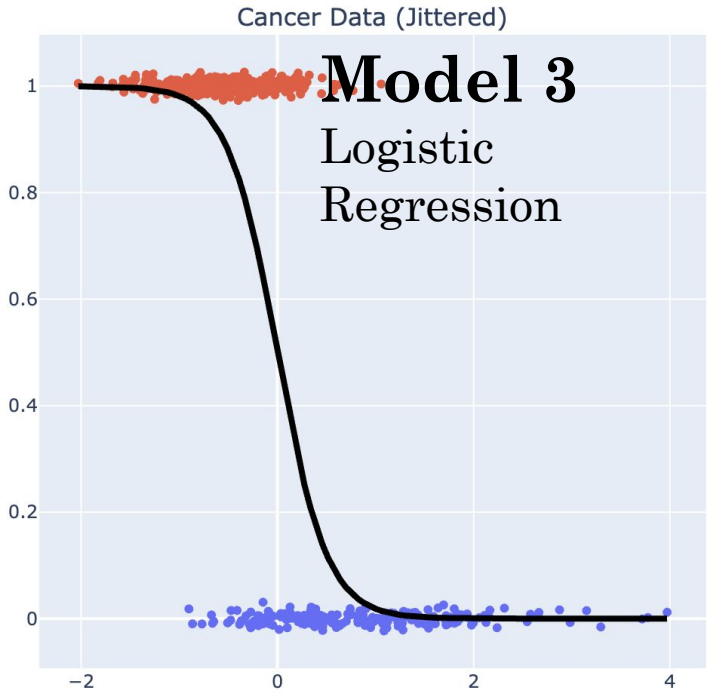
Demo

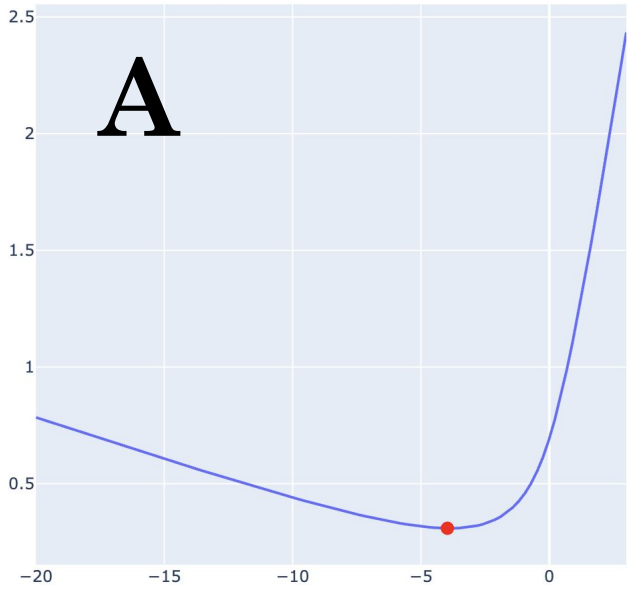
Understanding the Error Surface

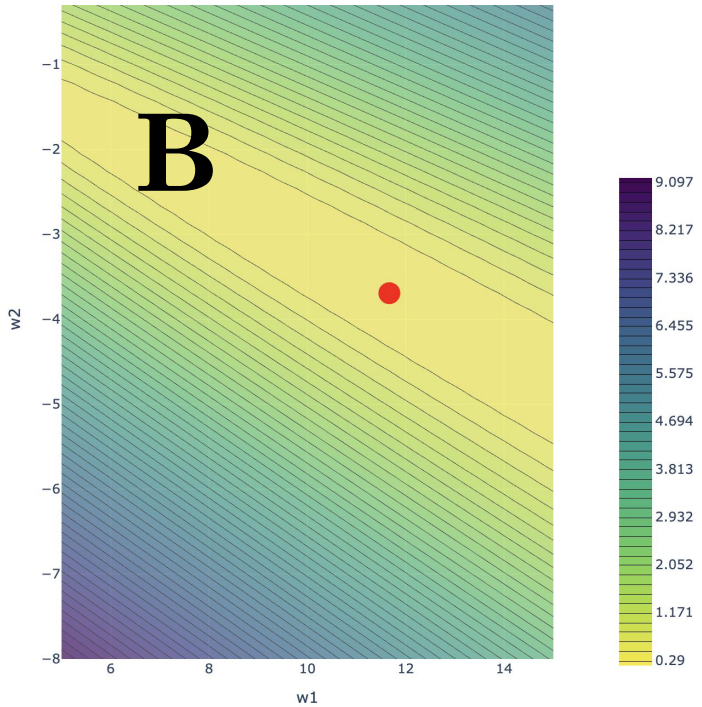


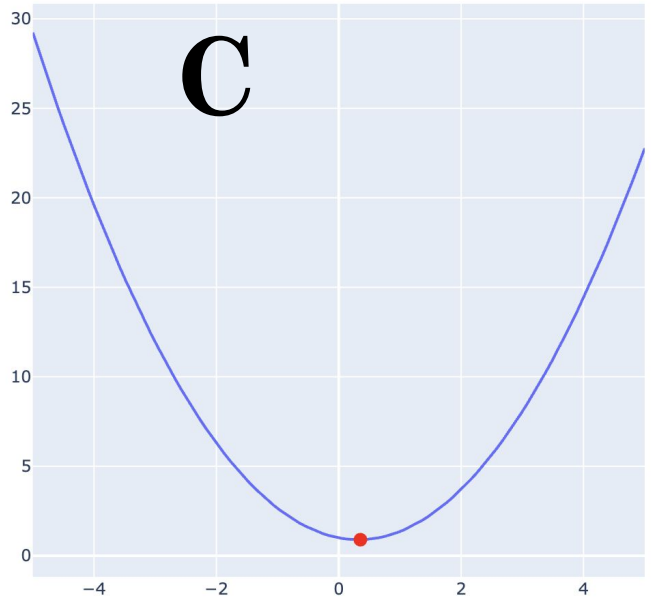

Model 1
Logistic
Regression
y
x


Cancer Data
Model 2
Least Squares
Regression


Cancer Data (Jittered)
Model 3
Logistic
Regression


A


B
w1
w2
9.097
8.217
7.336
6.455
5.575
4.694
3.813
2.932
2.052
1.171
0.29


C

Match the model with the loss.

Calculus and Optimization

The Gradient

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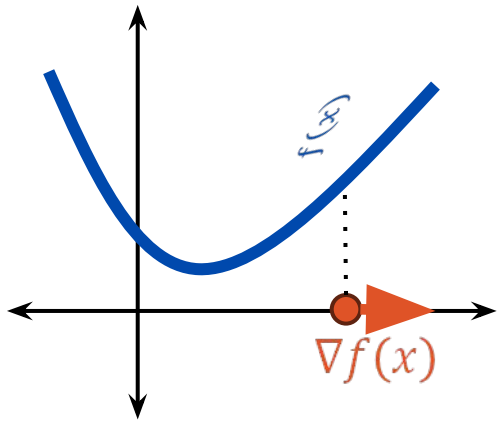
The gradient of a **scalar-valued function** $f: \mathbb{R}^d \to \mathbb{R}$ is the **column vector-valued function** $\nabla f: \mathbb{R}^d \to \mathbb{R}^d$ of partial derivatives of f

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix}$$



Important Properties

- The gradient $\nabla f(x)$ points in the **direction of steepest ascent**.
- The gradient $\nabla f(x) = 0$ at local minima, maxima, and saddle points.
  - We already used this property to solve for the MLE (…many times…)

Gradient Exercise

Gradient

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix}$$

Consider the function:

$$E(w) = (w_0 - 1)^2 + (w_1 - 2)^2 + 1$$

Compute the partial derivatives:

$$\frac{\partial}{\partial w_0} E(w) = 2(w_0 - 1)$$

$$\frac{\partial}{\partial w_1} E(w) = 2(w_1 - 2)$$

Then the gradient is:

$$\nabla E(w) = \begin{bmatrix} 2(w_0 - 1) \\ 2(w_1 - 2) \end{bmatrix}$$

Gradient

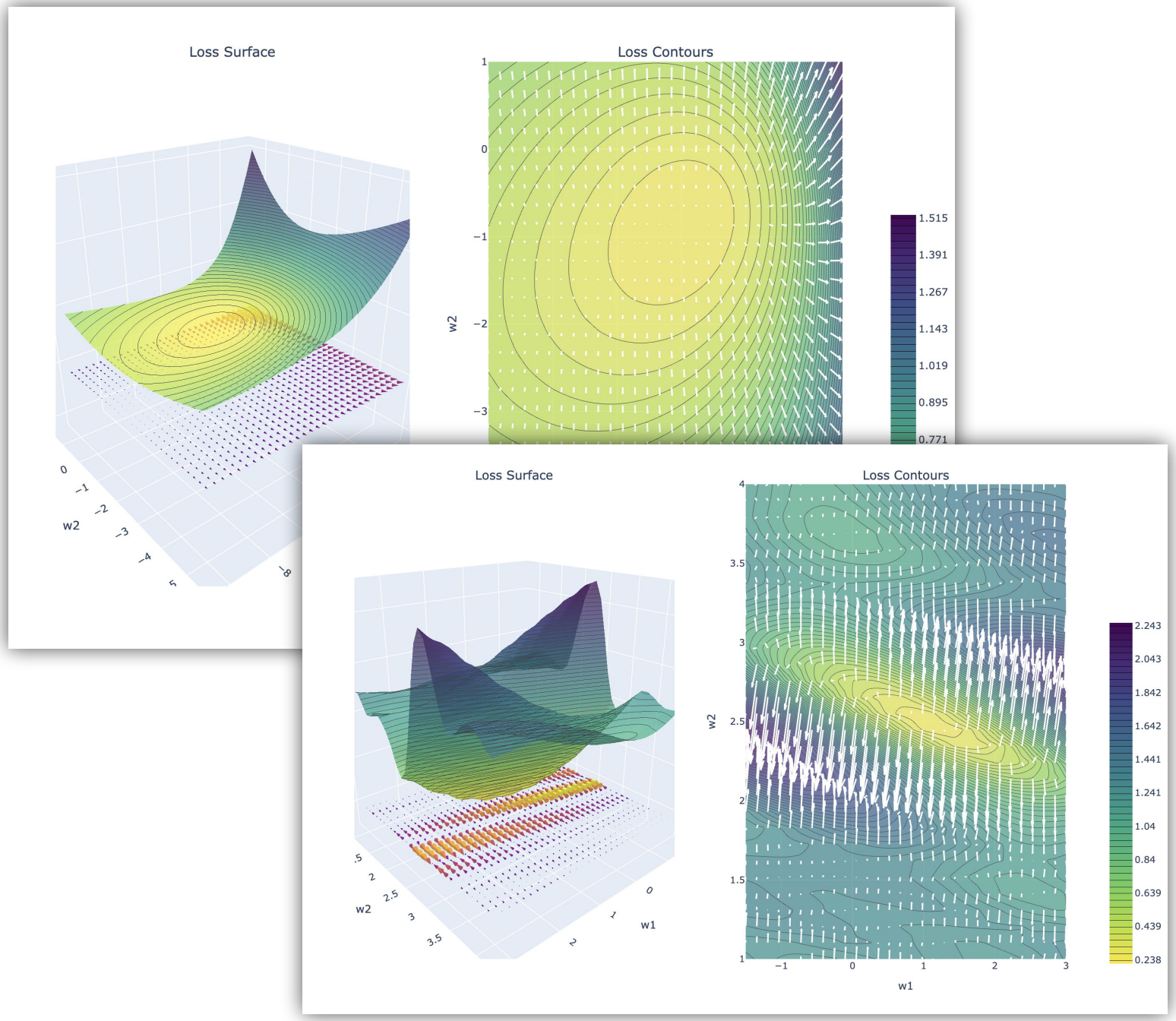
The gradient points in the **direction of greatest ascent**.

Zero Gradient at $w = (1,2)$

Plots in demo notebook.

Demo

The Gradient of the Error Function



If w is a d-dimensional vector and E(w) is a scalar-valued function, what is the dimensionality of the gradient $\nabla E(w)$ with respect to w?

Common Confusion

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$$E(w) = (w_0 - 1)^2 + (w_1 - 2)^2 + 1$$

- The gradient does not lie on the surface of the function.

Not the gradient!

Example: $E: \mathbb{R}^2 \to \mathbb{R}$

The gradient $\nabla E: \mathbb{R}^2 \to \mathbb{R}^2$ is **2-dimensional (not 3)**

The gradient is in the **domain** of the function.

Stationary Points and Zero Gradient

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Anywhere the gradient is zero is a **stationary point.**

- Gradient is zero at local minima but zero gradient does not imply a minima

$$E_A[w] = (w_0 - 1)^2 + (w_1 - 2)^2 + 1$$

Minimum

$$E_B[w] = -(w_0 - 1)^2 - (w_1 - 2)^2 + 12$$

Maximum

$$E_C[w] = -(w_0 - 1)^2 + (w_1 - 2)^2 + 1$$

Saddle Point

Plots in demo notebook.

Analytical Optimization using Gradients

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- In some situations, (e.g., **ordinary least squares**) we can find the stationary points by solving:

$$\nabla E(w) = 0$$

- For these problems we often know that there is one stationary point and that it is the minima.

However, for many machine learning applications (e.g., logistic regression and neural networks) **we cannot solve for the stationary points analytically**.

Iterative optimization

Gradient Descent

Iterative Optimization Algorithms

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Iterative optimization algorithms start with an **initial estimate** $\mathbf{w}^{(0)}$ and then **iteratively refine** that estimate:

$$\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} + \Delta\mathbf{w}^{(\tau-1)}$$

- Iteration τ
- New Estimate
- Old Estimate
- Update (Based on **Gradient**)

until **stopping criterion** is achieved.

The choice of initial estimate $\mathbf{w}^{(0)}$, how the update $\Delta\mathbf{w}^{(\tau-1)}$ is computed, and the **stopping criterion** define the **iterative optimization algorithm**.

Intuition

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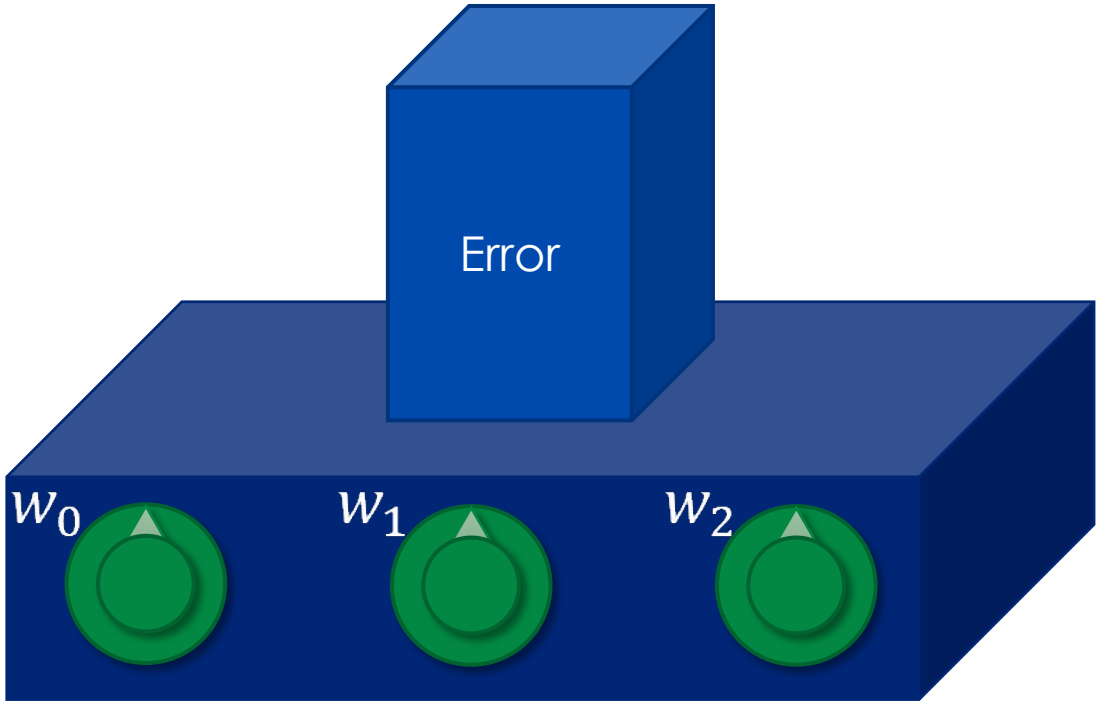

Goal: Minimize the loss by turning the knobs.

Try the loss game (its free)!

Intuition

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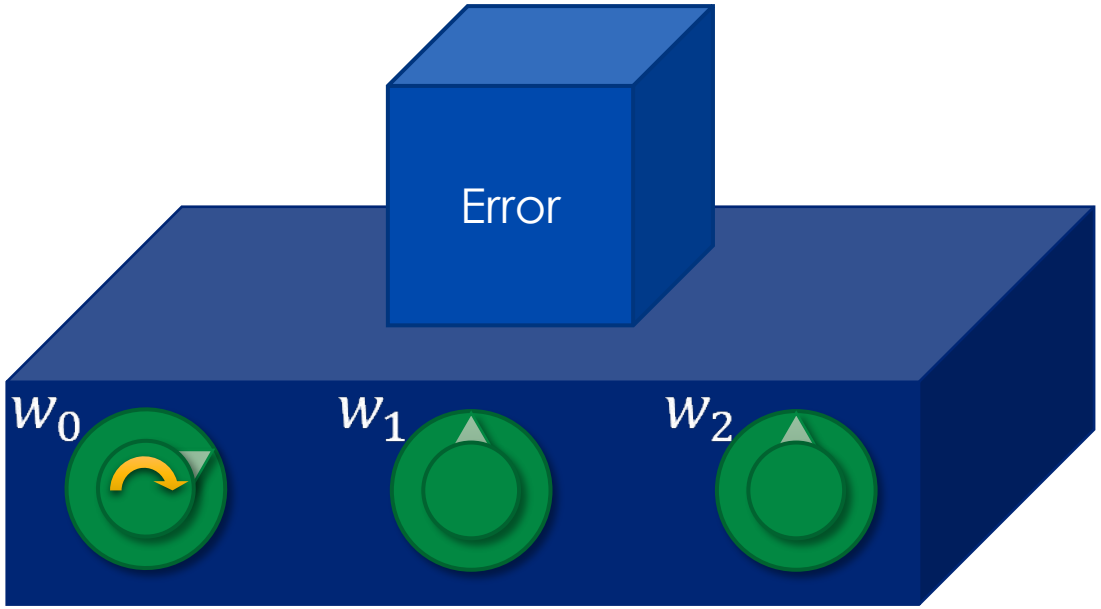
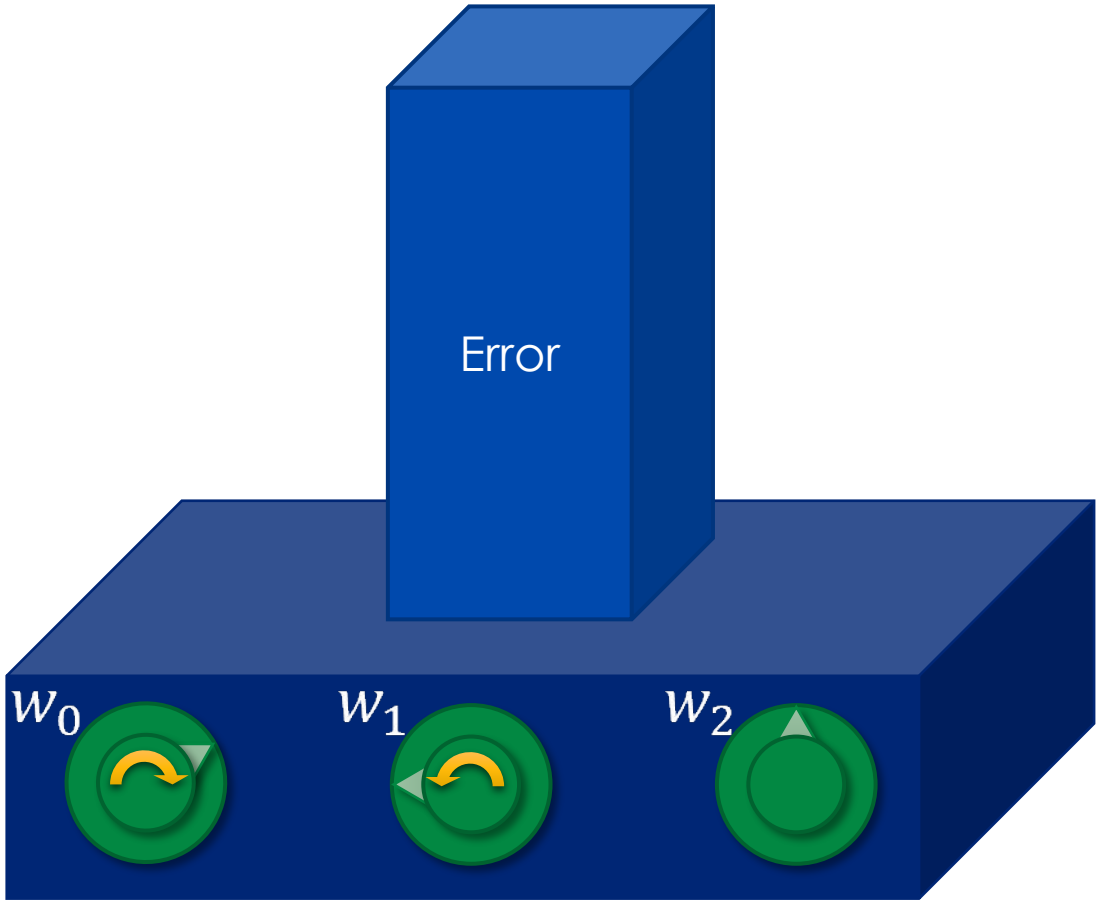


Try the loss game (its free)!

Intuition



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Try the loss game (its free)!

Intuition

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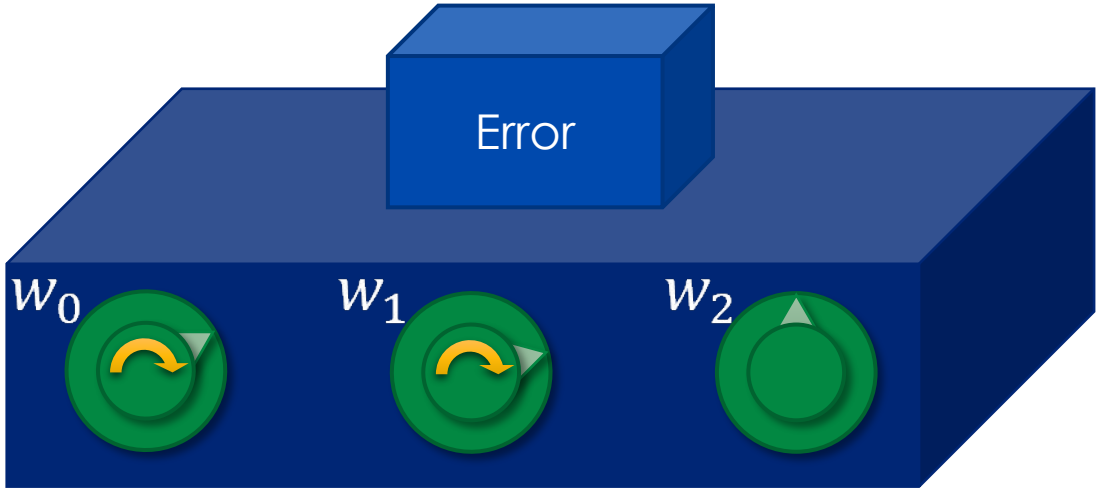

Try the loss game (its free)!

Intuition

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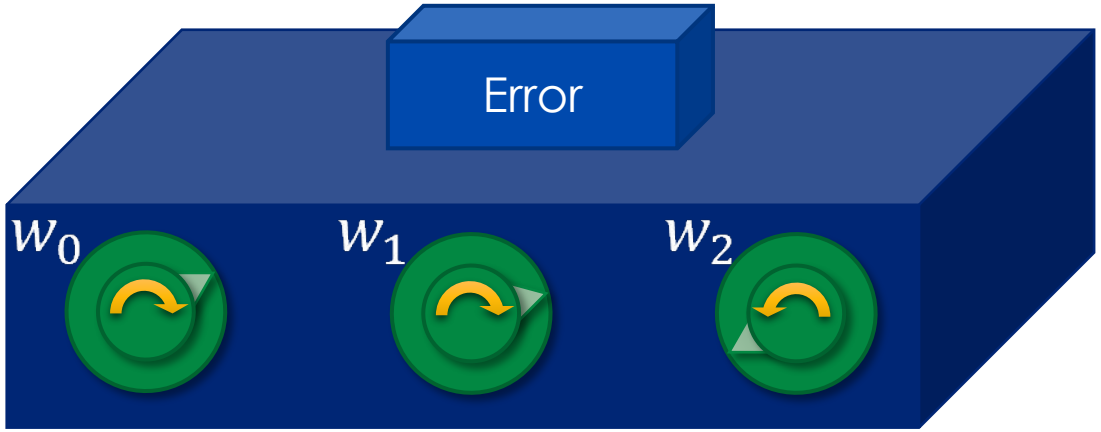

Try the loss game (its free)!

Intuition

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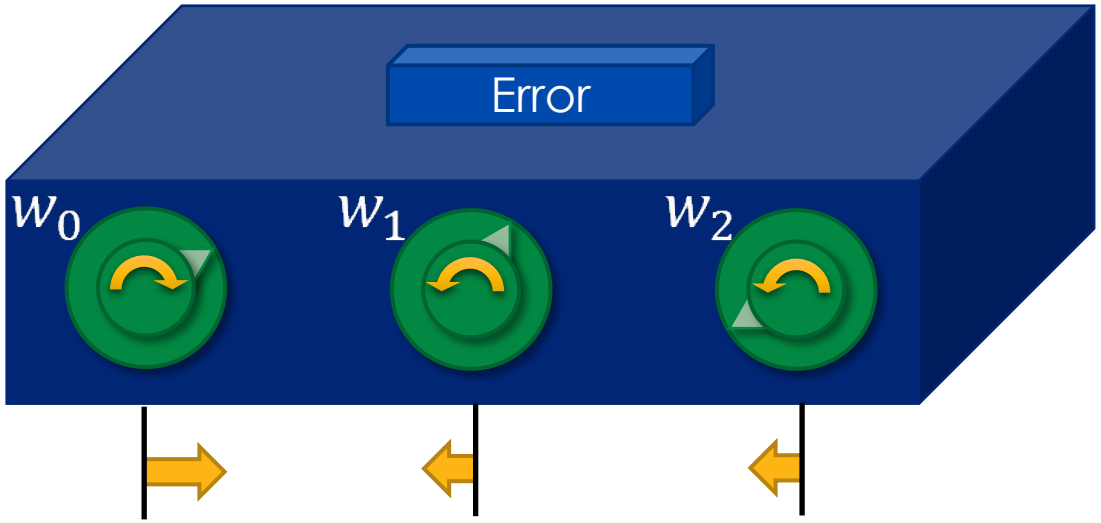

What if we knew which way to turn the knob and an idea of how far?

This is the Gradient!

Try the loss game (its free)!

Intuition

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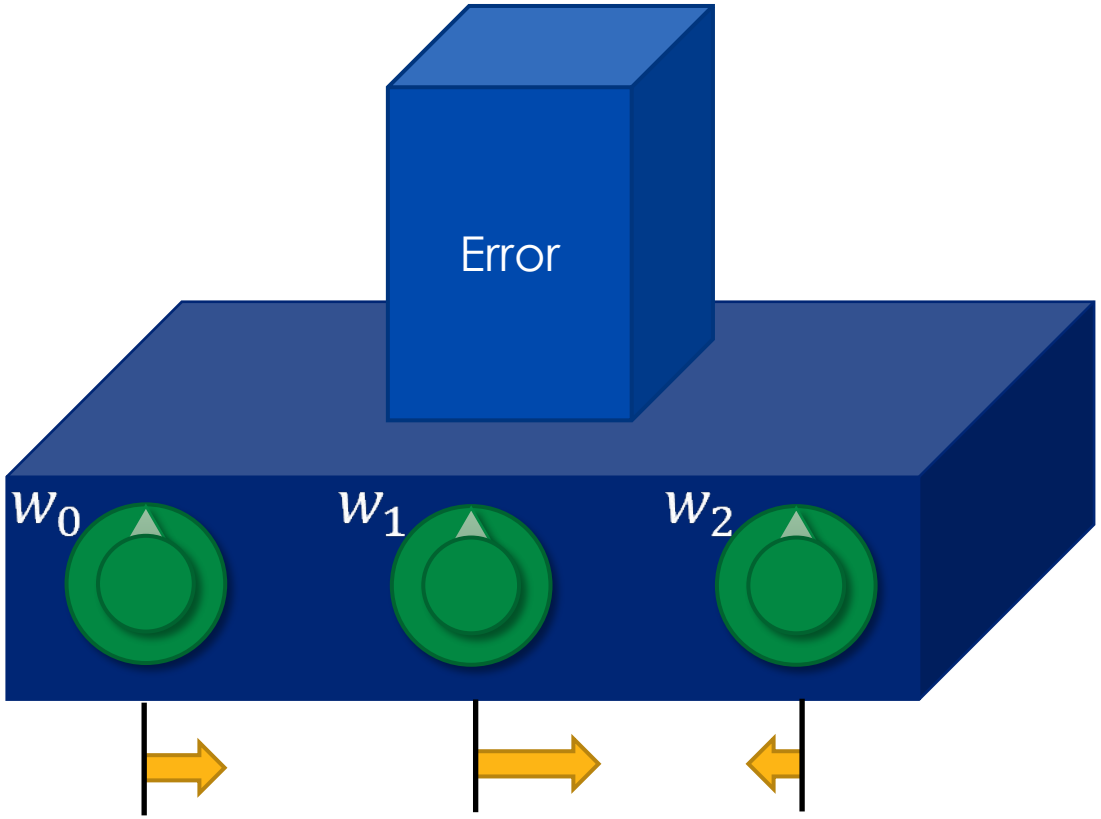

Try the loss game (its free)!

Intuition

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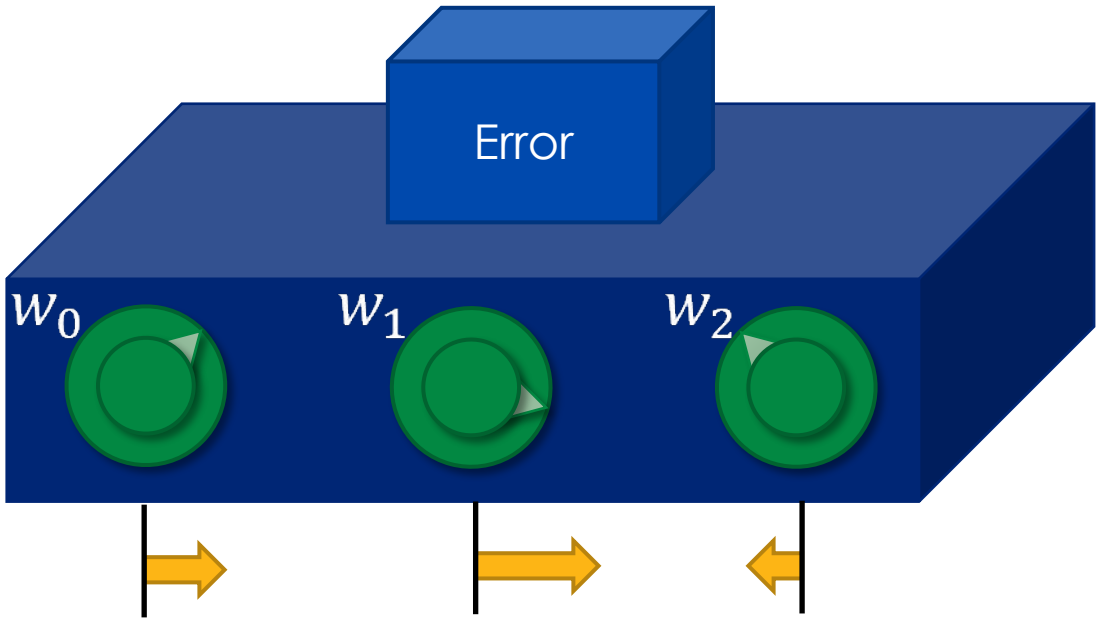

Try the loss game (its free)!

Intuition

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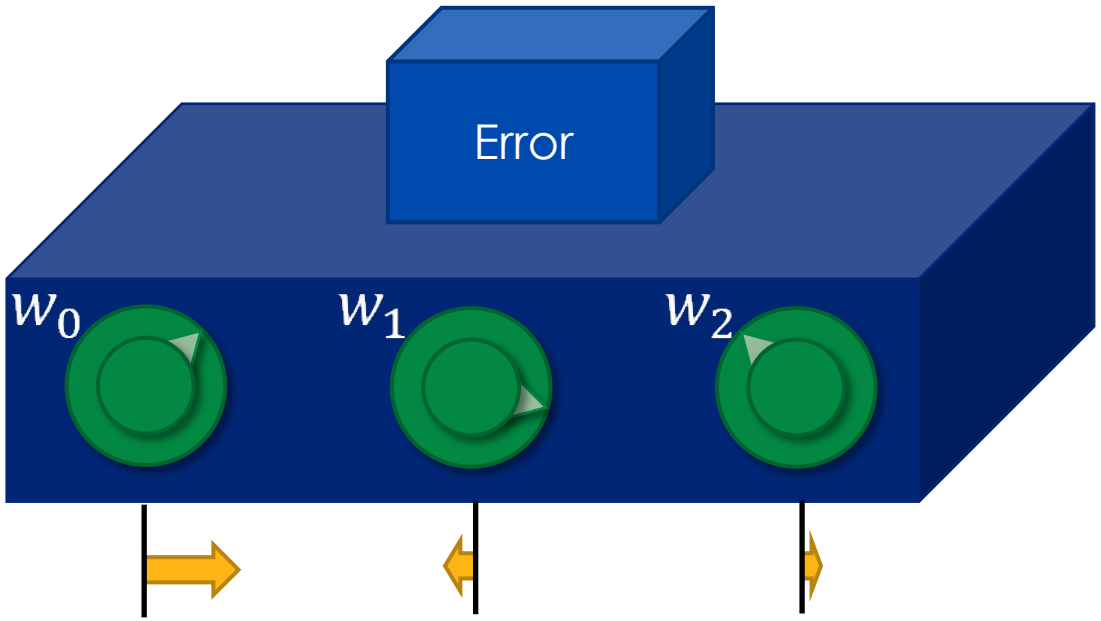

Try the loss game (its free)!

Intuition

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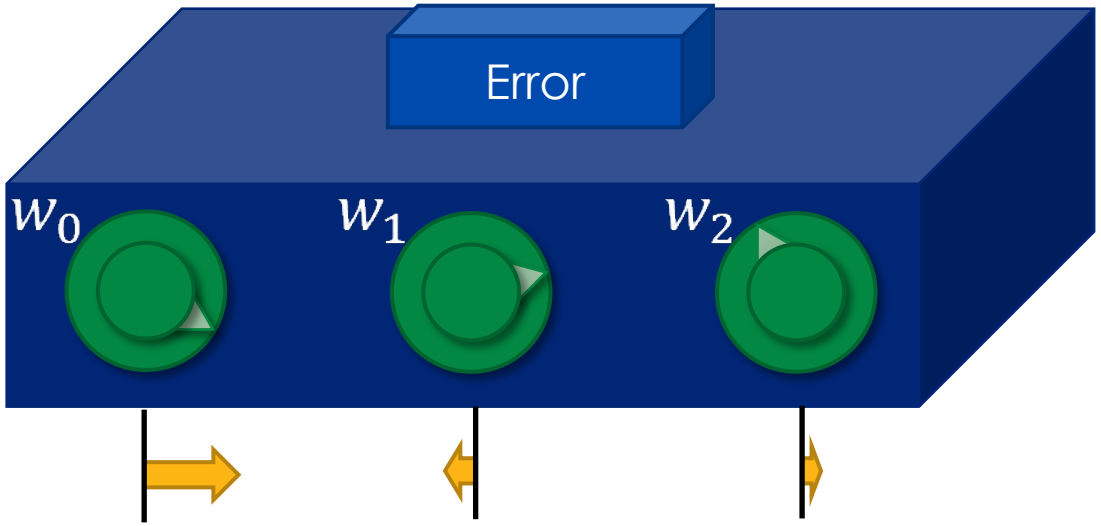

Try the loss game (its free)!

Intuition

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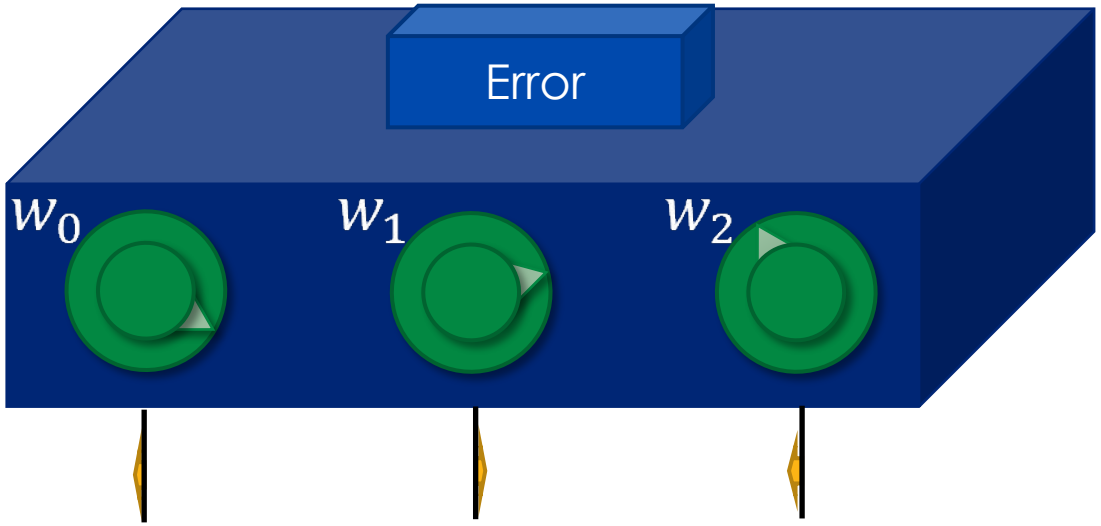

This is Gradient descent!

Try the loss game (its free)!

Gradient Descent Intuition

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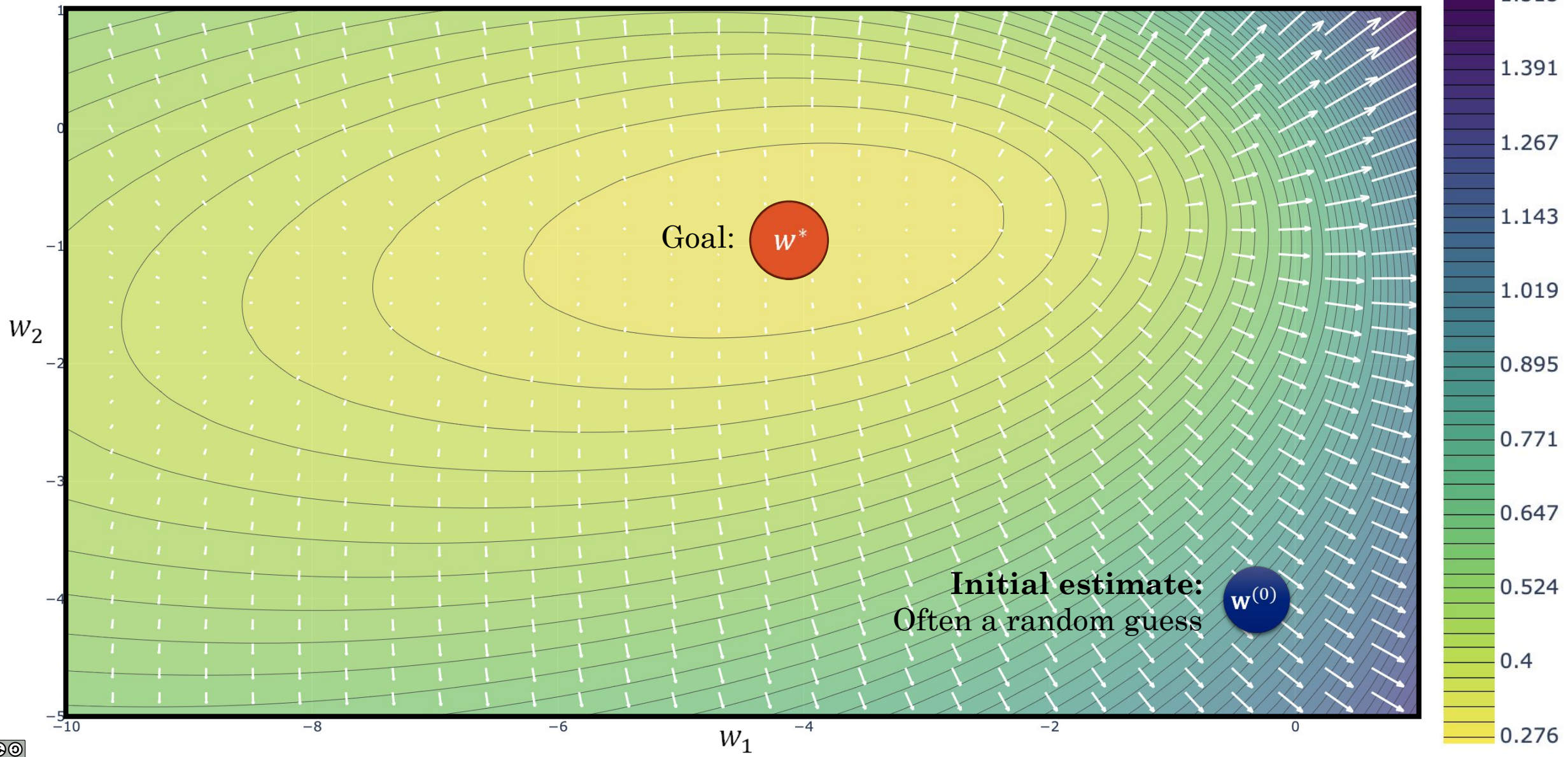

Gradient Descent Intuition

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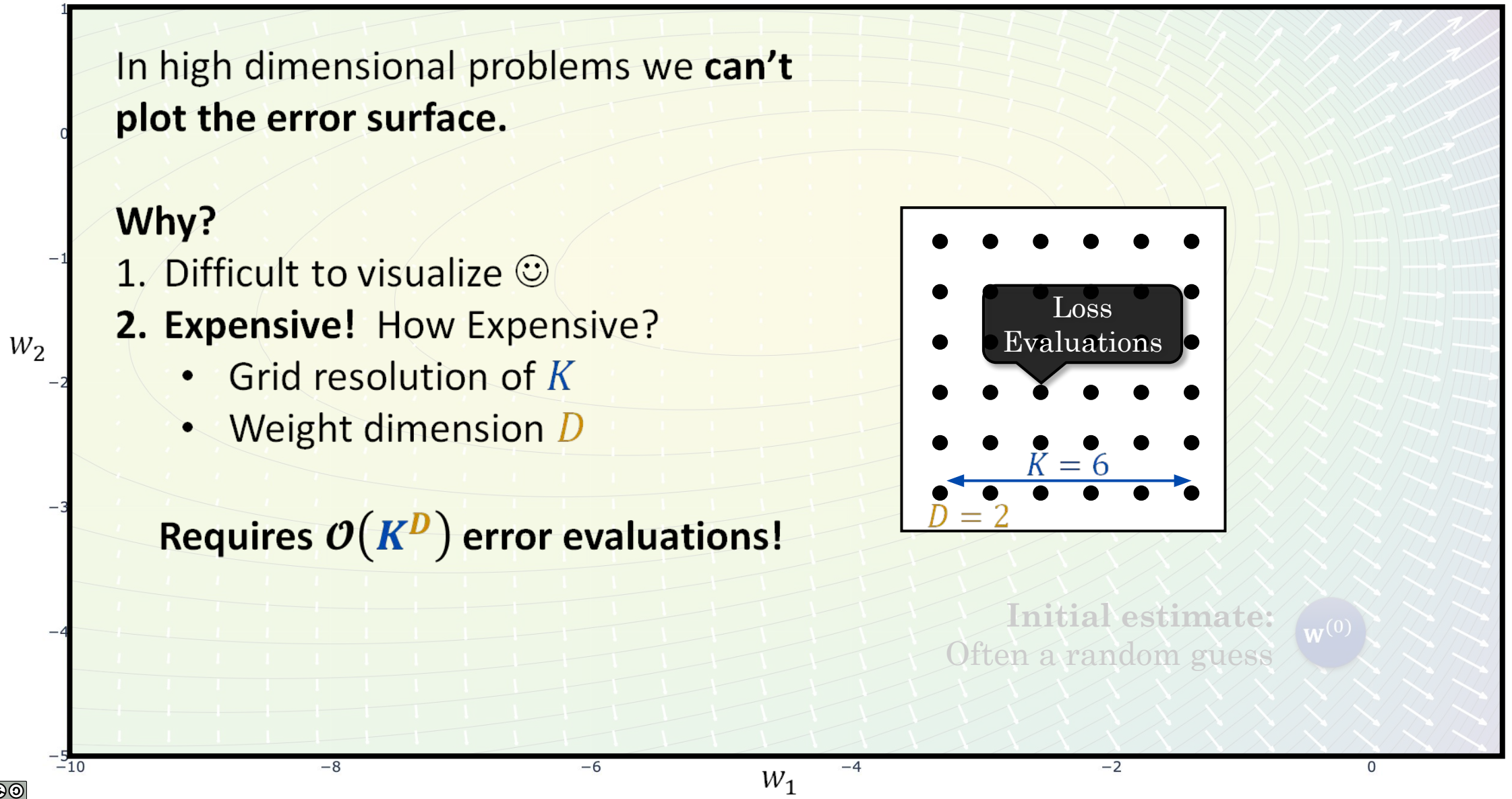

Gradient Descent Intuition

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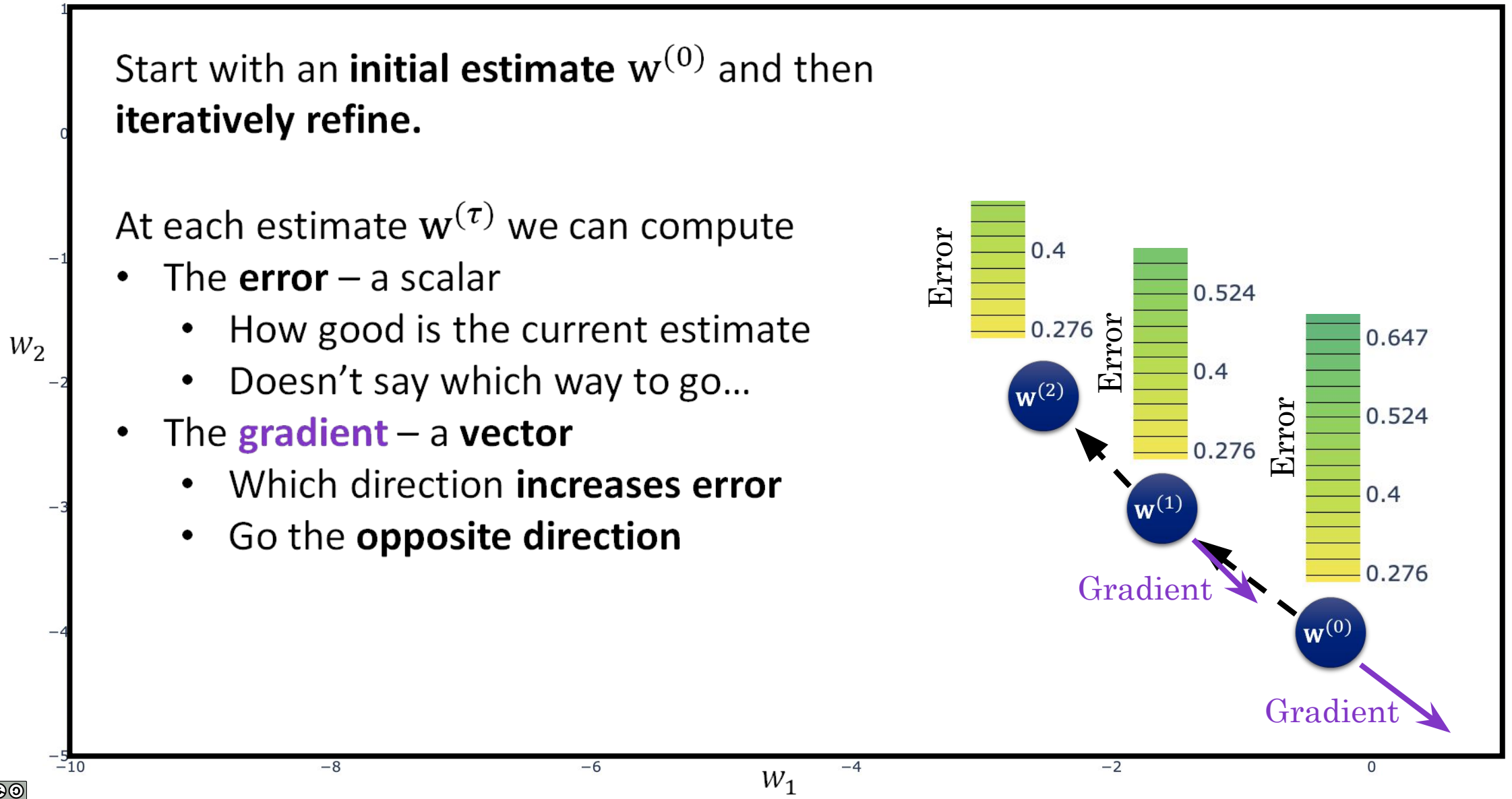
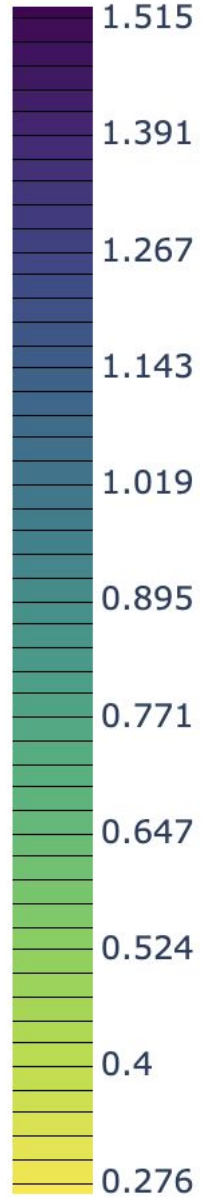
Start with an **initial estimate** $\mathbf{w}^{(0)}$ and then **iteratively refine.**

At each estimate $\mathbf{w}^{(\tau)}$ we can compute

- The **error** – a scalar
  - How good is the current estimate
  - Doesn't say which way to go…
- The **gradient** – a **vector**
  - Which direction **increases error**
  - Go the **opposite direction**





The (Batch) **Gradient Descent** Alg.

Update the weights by moving in the **opposite direction** of the gradient:

$$\mathbf{w}^{(0)} = \text{Initialize}()$$
$$\text{for } \tau \text{ in range}(1, \tau_{\max}):$$
$$\quad \mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} - \eta \nabla E\left(\mathbf{w}^{(\tau-1)}\right)$$
$$\quad \text{if } \text{Converged}\left(\mathbf{w}^{(\tau)}, \mathbf{w}^{(\tau-1)}, \epsilon\right): \text{terminate}$$

$\text{Initialize}()$ typically a small random value (more on this later)

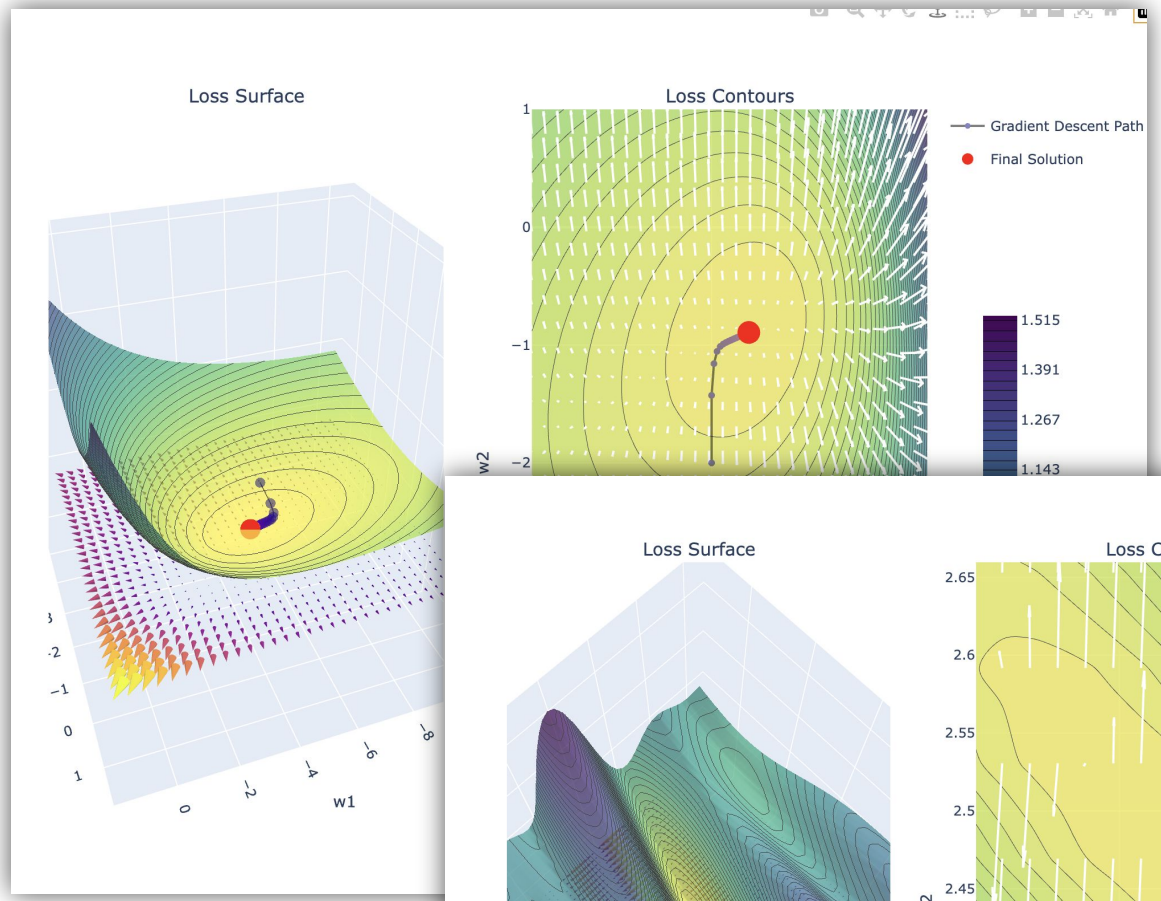
$\text{Converged}\left(\mathbf{w}^{(\tau)}, \mathbf{w}^{(\tau-1)}, \epsilon\right)$ stopping condition (e.g., ϵ change in $\Delta\mathbf{w}^{(\tau-1)}$)

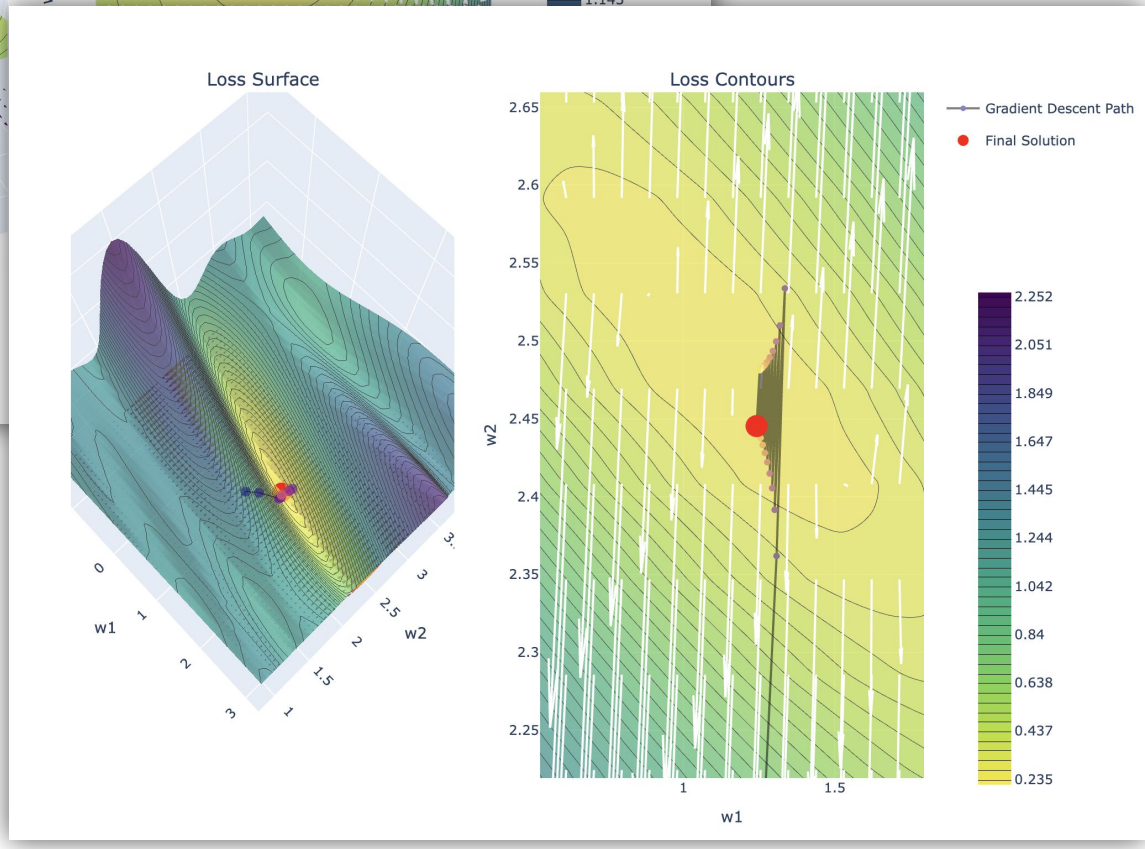
Hyper Parameters:

- η is the learning rate (typically a small value $0 < \eta < 1$, more on this later)
- $\tau_{\max}$ is the maximum number of iterations

Demo

Batch Gradient Descent





Loss (Error) Curves

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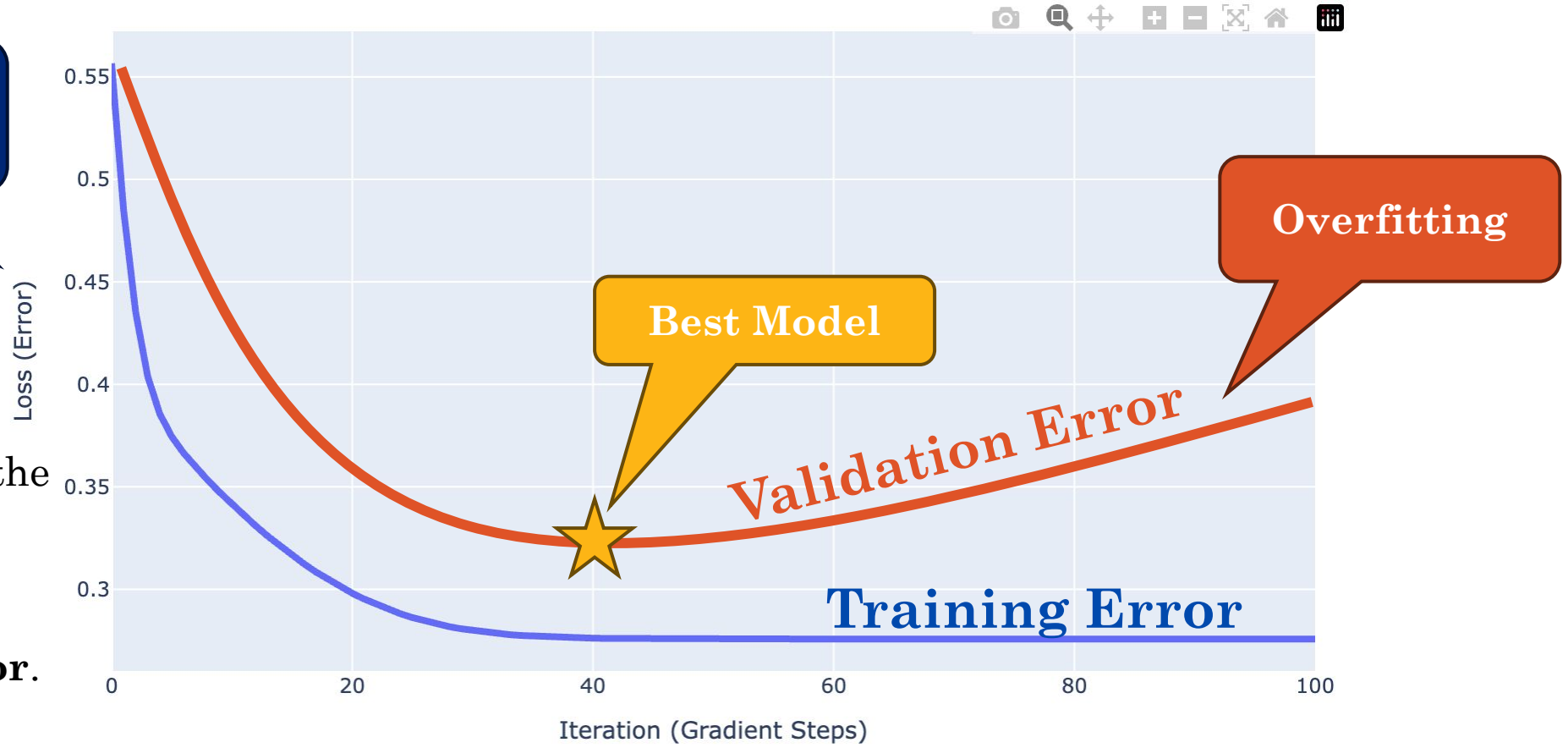
The **loss** or **error curve plots** the error as a function of the number of **iterations** (**steps**) of gradient descent.

Training and **Validation** Error

Validation curves are typically **above** the training curves.

You can **overoptimize** the objective (**overfit**).

Choose parameters with **lowest validation error**.



Convergence Assuming Quadratic Approximation

Assuming a simple quadratic equation (e.g., x^2) is gradient descent guaranteed to converge?

Quadratic Approximations

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- The 2nd order (quadratic) Taylor approximation of a function function $f: \mathbb{R} \to \mathbb{R}$ evaluated at $\hat{x}$ is:

$$\tilde{f}(x) = f(\hat{x}) + \left(\frac{\partial}{\partial x} f(x)\Big|_{x=\hat{x}}\right)(x - \hat{x}) + \frac{1}{2}\left(\frac{\partial^2}{\partial x^2} f(x)\Big|_{x=\hat{x}}\right)(x - \hat{x})^2$$

For example: $f(x) = \sin(x)$

$$\tilde{f}_{\hat{x}=1}(x) = \sin(1) + \cos(1)\,(x-1) + \frac{1}{2}(-\sin(1))(x-1)^2$$

$$\tilde{f}_{\hat{x}=5}(x) = \sin(5) + \cos(5)\,(x-5) + \frac{1}{2}(-\sin(5))(x-5)^2$$

Taylor Expansion of the Error Surface

The 2nd-order Taylor expansion of the **error surface** at $\hat{w}$:

$$\tilde{E}_{\hat{w}}(w) = E(\hat{w}) + (w - \hat{w})^T b + \frac{1}{2}(w - \hat{w})^T H (w - \hat{w})$$

- $b = \nabla E(\hat{w})$ is the **Gradient** of E evaluated at $\hat{w}$ (vector)
- $H = \nabla\nabla E(\hat{w})$ is the **Hessian** of E evaluated at $\hat{w}$ (a matrix)

Matrix Equivalent of the Second Derivative

Hessians (the 2nd derivative)

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The Hessian of a **scalar-valued function** $E: \mathbb{R}^D \to \mathbb{R}$ is the **matrix-valued function** $\nabla^2 E: \mathbb{R}^D \to \mathbb{R}^{D \times D}$ of the 2nd partial derivatives of f

$$H = \nabla^2 E = \begin{bmatrix} \frac{\partial^2}{\partial x_1^2} E(x) & \dots & \frac{\partial^2}{\partial x_1 x_D} E(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2}{\partial x_D x_1} E(x) & \dots & \frac{\partial^2}{\partial x_D^2} E(x) \end{bmatrix}$$

The Hessian

- describes the local **curvature** of E
- determines the **critical point type** (minima, maxima, or saddle)

Stopped Here

Will return to later material in the next lecture.

Hessian Exercise

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Consider the function:

$$E(w) = (w_0 - 1)^2 + (w_1 - 2)^2 + 1$$

Then the gradient is: $\nabla E(w) = \begin{bmatrix} 2(w_0 - 1) \\ 2(w_1 - 2) \end{bmatrix}$

The Hessian is:

$$\mathrm{H(w)} = \begin{bmatrix} \frac{\partial^2}{\partial w_0^2} E(w) & \frac{\partial^2}{\partial w_0 \partial w_1} E(w) \\ \frac{\partial^2}{\partial w_0 \partial w_1} E(w) & \frac{\partial^2}{\partial w_1^2} E(w) \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Hessian

$$\nabla^2 E = \begin{bmatrix} \frac{\partial^2}{\partial x_1^2} E(x) & \cdots & \frac{\partial^2}{\partial x_1 x_D} E(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2}{\partial x_D x_1} E(x) & \cdots & \frac{\partial^2}{\partial x_D^2} E(x) \end{bmatrix}$$

Plots in demo notebook.

Taylor Expansion of the Error Surface

The 2nd-order Taylor expansion of the error surface at $\hat{w}$:

$$\tilde{E}_{\hat{w}}(w) = E(\hat{w}) + (w - \hat{w})^T b + \frac{1}{2}(w - \hat{w})^T H (w - \hat{w})$$

- $b = \nabla E(\hat{w})$ is the **Gradient** of E evaluated at $\hat{w}$ (vector)
- $H = \nabla\nabla E(\hat{w})$ is the **Hessian** of E evaluated at $\hat{w}$ (a matrix)

If $\hat{w}$ is a **stationary point** (w^*) then $\nabla E(w^*) = 0 = b$

$$\tilde{E}_{w^*}(w) = E(w^*) + \frac{1}{2}(w - w^*)^T H (w - w^*)$$

Use a change of variables to better understand this part.

Eigen Decomposition of the Hessian

We can construct the **Eigen decomposition of the Hessian**

$$Hu_i = \lambda_i u_i$$

- u_i are the complete set of **orthonormal eigen vectors:** $\forall_{i,j}: u_i^T u_i = \delta_{ij}$
- λ_i are the real-valued **eigen values**

Then we apply a **change of variables**:

$$w - w^* = \sum_i \alpha_i u_i \quad \text{where} \quad \alpha_i = (w - w^*)^T u_i$$

This allows us to rewrite:

$$\tilde{E}_{w^*}(w) = E(w^*) + \frac{1}{2}(w - w^*)^T H (w - w^*) = E(w^*) + \frac{1}{2}\sum_i \alpha_i^2 \lambda_i$$

Show it!

- u_i are the complete set of **orthonormal eigen vectors:** $\forall_{i,j}\colon\ u_i^T u_i = \delta_{ij}$
- λ_i are the real-valued **eigen values**

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This allows us to rewrite:

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Show it!

Derivation:

$$(w - w^*)^T H (w - w^*) = \left(\sum_i \alpha_i u_i\right)^T H \left(\sum_i \alpha_i u_i\right) = \left(\sum_i \alpha_i u_i\right)^T \left(\sum_i \alpha_i H u_i\right)$$

$$= \left(\sum_i \alpha_i u_i\right)^T \left(\sum_i \alpha_i \lambda_i u_i\right) = \sum_{ij} \alpha_i \alpha_j u_i^T u_j \lambda_j = \sum_{ij} \alpha_i \alpha_j \delta_{ij} \lambda_j = \sum_i \alpha_i^2 \lambda_i$$

Eigenvector Defn.

Orthonormality

Taylor Expansion at the Stationary Pts.

If $\hat{w}$ is a **stationary point** (w^*) (where $b = \nabla E|_{w=\hat{w}} = 0$)

$$\tilde{E}_{w^*}(\alpha) = E(w^*) + \frac{1}{2}\sum_i \alpha_i^2 \lambda_i$$

At $\alpha = 0$ we get the error at the critical point $E(w^*)$.

- If we move in **direction *i*** (corresponding to u_i) then α_i^2 **increases.**
  - The change in error then depends on the sign of λ_i

Therefore, if the **eigenvalues are**

- **all positive** then w^* is a **local minimum.**
- **all negative** then w^* is a **local maximum.**
- **mixed** then w^* is a **saddle point.**

Stationary Points and the Hessian

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Eigenvalues of the Hessian determine the curvature at the critical points

Minimum	**Maximum**	**Saddle Point**

Eigenvector of the Hessian

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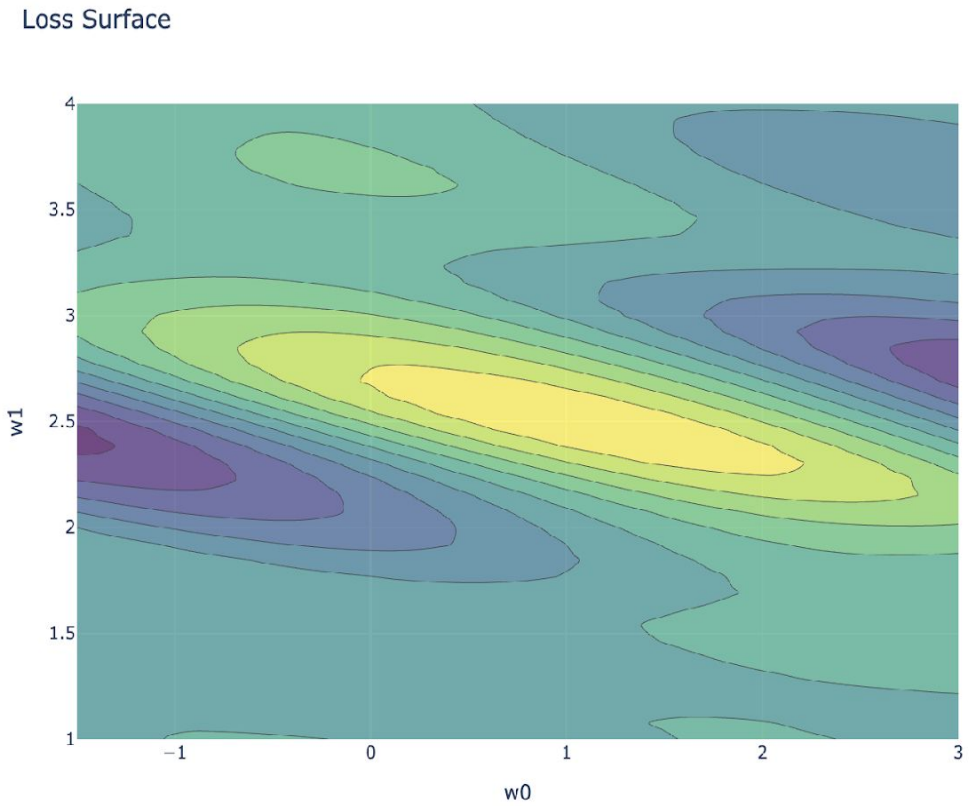
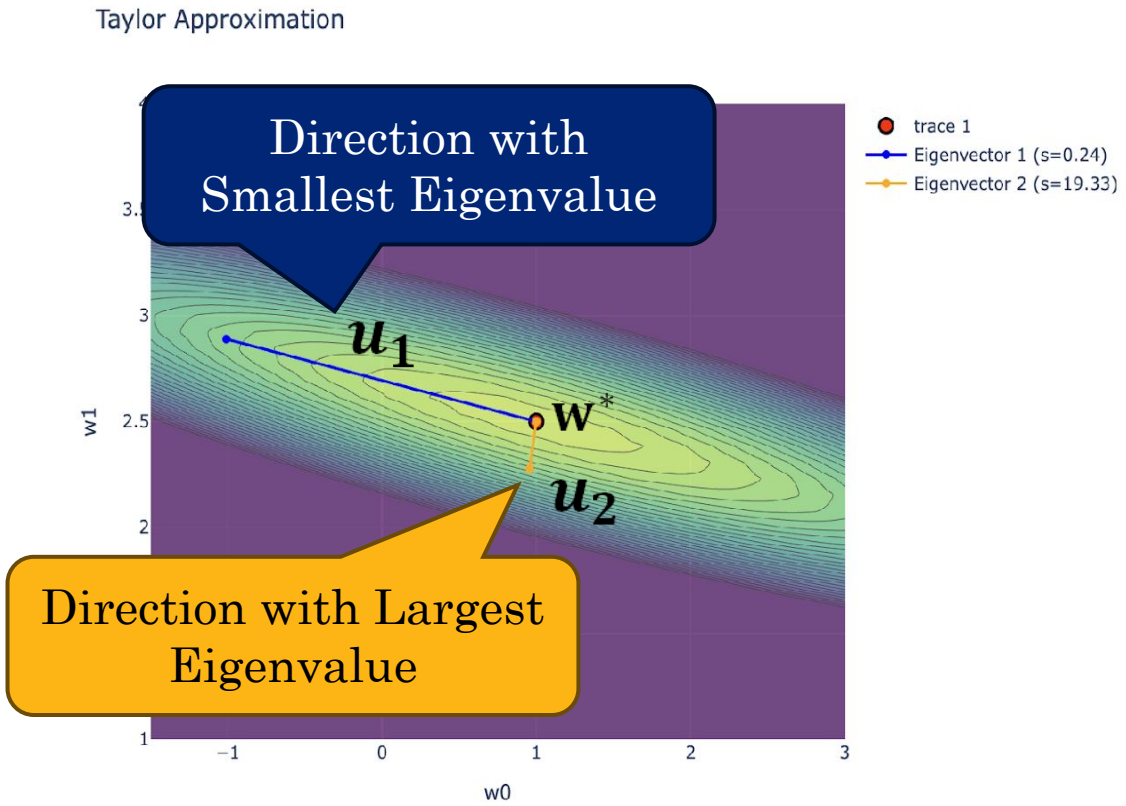
Elliptical contours of constant error align with the eigenvectors of the Hessian matrix.

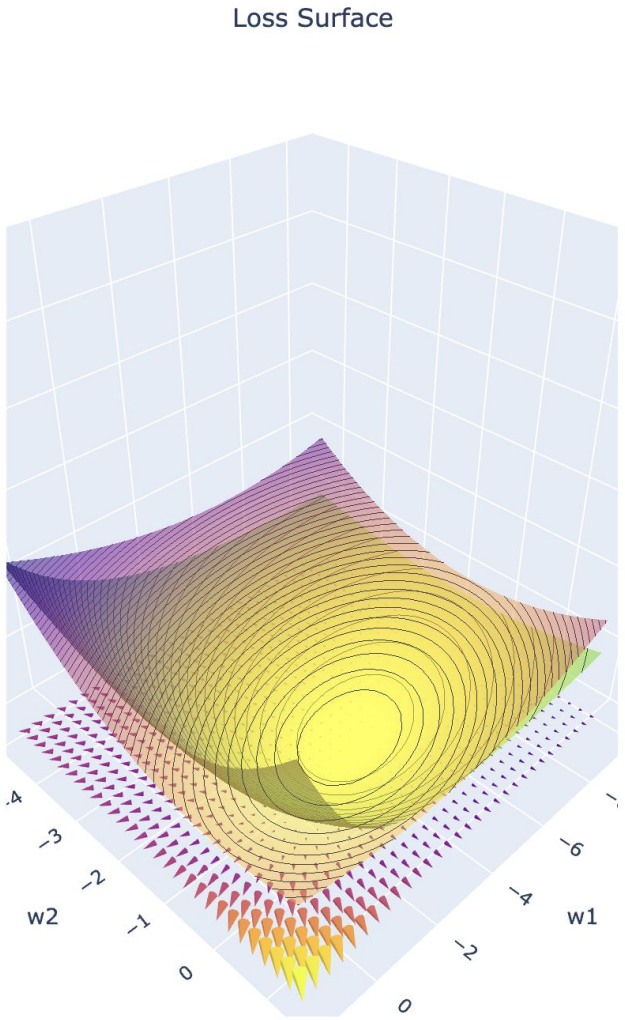


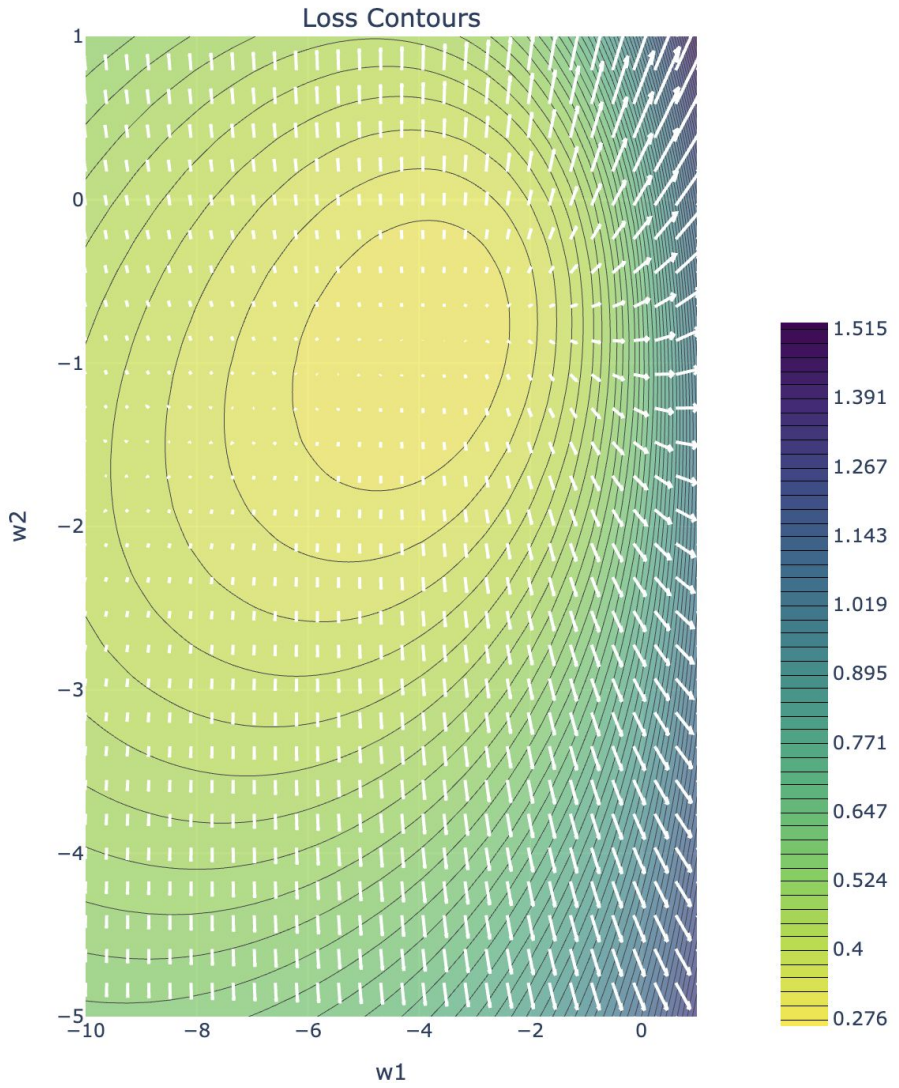


Plots in demo notebook.

Demo

Taylor Expansions and the Hessian





Recap: Quadratic Approx. Error

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We constructed the **2nd-order Taylor expansion** of the error surface at the **stationary point** w^* (where $b = \nabla E|_{w=w^*} = 0$):

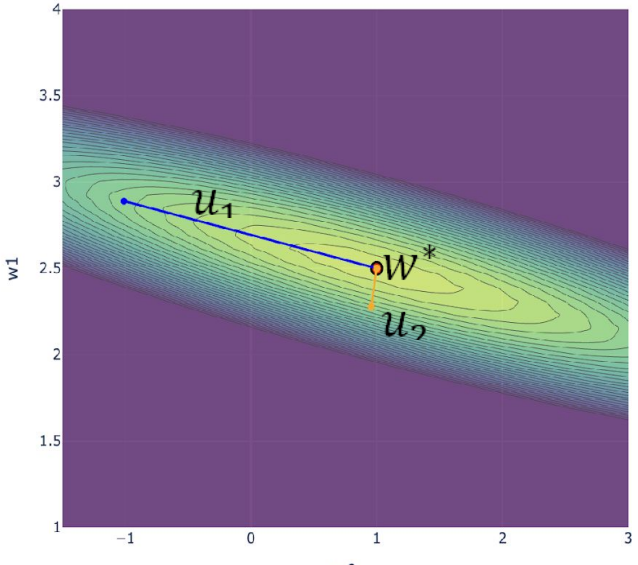
$$\tilde{E}_{w^*}(w) = E(w^*) + \frac{1}{2}(w - w^*)^T H (w - w^*)$$

Using the **Eigen-decomposition of the Hessian:** $Hu_i = \lambda_i u_i$ we applied a change of variables $\alpha_i = (w - w^*)^T u_i$ to obtain:

$$\tilde{E}_{w^*}(\alpha) = E(w^*) + \frac{1}{2}\sum_i \alpha_i^2 \lambda_i$$

- **This is minimized at $\alpha = 0$**

We can **evaluate gradient descent** on this simple quadratic approximation to better understand the convergence properties.



Grad. Descent on the Quadratic Approx.

Quadratic Approx. Error

$$\tilde{E}_{w^*}(\alpha) = E(w^*) + \frac{1}{2}\sum_i \alpha_i^2 \lambda_i$$

- Compute the **gradient** with respect to α_i:

$$\frac{\partial}{\partial \alpha_i} \tilde{E}_{w^*}(\alpha) = \alpha_i \lambda_i$$

Which gives the **gradient descent update**:

$$\alpha_i^{(\tau)} = \alpha_i^{(\tau-1)} - \eta\left(\alpha_i^{(\tau-1)} \lambda_i\right) = (1 - \eta\lambda_i)\alpha_i^{(\tau-1)}$$

If we **iteratively** apply the gradient descent update, we obtain:

$$\alpha_i^{(1)} = (1 - \eta\lambda_i)\alpha_i^{(0)} \rightarrow \alpha_i^{(2)} = (1 - \eta\lambda_i)\left((1 - \eta\lambda_i)\alpha_i^{(0)}\right) \rightarrow \ldots \rightarrow$$

$$\alpha_i^{(\tau)} = (1 - \eta\lambda_i)^{\tau}\alpha_i^{(0)}$$

Grad. Descent on the Quadratic Approx.

Assuming a **quadratic approximation** to the error function around the minima, the τ^{th} iteration of gradient descent we have:

$$\alpha_i^{(\tau)} = (1 - \eta\lambda_i)^{\tau}\alpha_i^{(0)}$$

and the **error is minimized when $\boldsymbol{\alpha_i = 0}$.**

To ensure $\alpha_i^{(\tau)} \to 0$ as $\tau \to \infty$ the learning rate should satisfy:

$$\boldsymbol{|1 - \eta\lambda_i| < 1}$$ **(Convergence Condition)**

- Smaller $\boldsymbol{|1 - \eta\lambda_i|}$ is better (converges faster)
- Negative $\boldsymbol{1 - \eta\lambda_i} < 0$ oscillatory behavior

Learning Rate Impact on Convergence

To maximize the **progress on each step** of gradient descent we want to **make η as large as possible (take bigger steps)**

- We want $(1-\eta\lambda_i)^\tau \to 0$ quickly as τ increases (larger η ➔ smaller $1-\eta\lambda_i$).
- However, to **ensure convergence we require $|1-\eta\lambda_i| < 1$ for all λ_i**

Let $\lambda_{\min}$ and $\lambda_{\max}$ be the smallest and largest **eigenvalues** of the Hessian

- Then $\eta < \frac{2}{\lambda_{\max}}$ because if $\eta \geq \frac{2}{\lambda_{\max}}$ then $|1-\eta\lambda_{\max}| \geq 1$

The **slowest converging dimension correspond to λ_{min}** so the smallest we can make $(1-\eta\lambda_i)$:

$$\left(1-\frac{2}{\lambda_{\max}}\lambda_{\min}\right) < (1-\eta\lambda_i)$$

Condition Number and Loss Surface

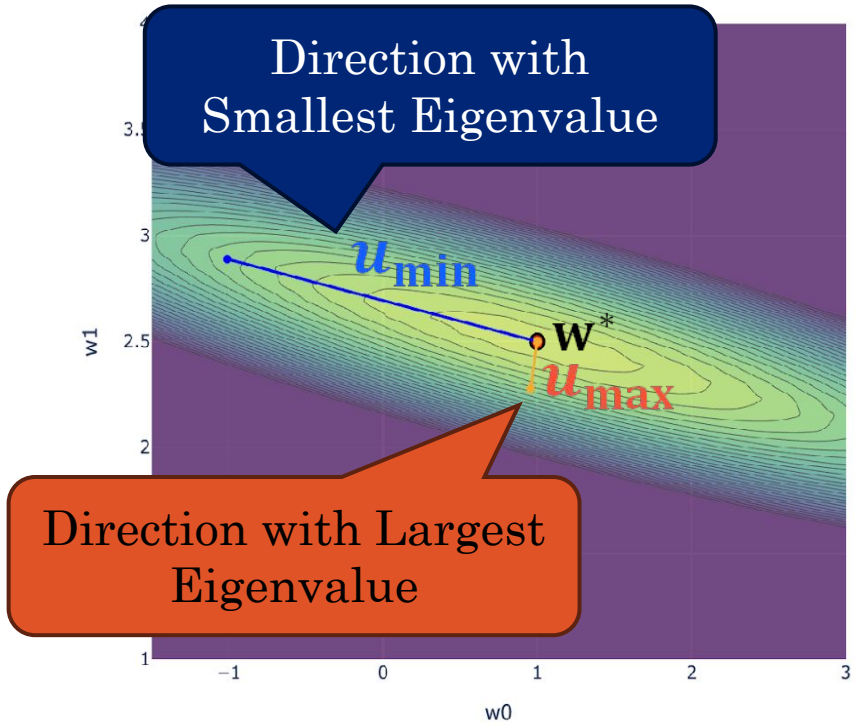
- The slowest converging dimension (i) is the dimension corresponding to λ_{min} and its convergence to 0 is lower bounded by:

$$\left(1 - \frac{2}{\lambda_{\max}} \lambda_{\min}\right)^{\tau} \alpha_i^{(0)}$$

The ratio of $\frac{\lambda_{\min}}{\lambda_{\max}}$ is the **condition number**.

- A problem is said to be **well conditioned** if it has a **large condition number.**

Gradient descent converges quickly on **well conditioned problems**.



Issues with Gradient Descent (GD)

GD converges slowly when the **error surface is relatively flat**:

Step size
(learning rate η) is
too small.

GD oscillates when the error surface is **poorly conditioned**

Step size
(learning rate η) is
too big.

Demo

Batch Gradient Descent

Momentum

Momentum can be used improve convergence in these settings by **accumulating common update directions**

$$\Delta \mathbf{w}^{(\tau-1)} = -\eta \nabla E\left(\mathbf{w}^{(\tau-1)}\right) + \mu \Delta \mathbf{w}^{(\tau-2)}$$

$$\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} + \Delta \mathbf{w}^{(\tau-1)}$$

- $0 \leq \mu < 1$ ($\mu = 0.9$) is the **momentum hyperparameter**

We now need to maintain the previous $\Delta \mathbf{w}^{(\tau-2)}$ update direction

Intuition: The solution **"picks up speed"** along the direction of repeated updates.

How Momentum Helps: Flat Directions

In **directions** that **are relatively flat** (e.g., is approx. constant ∇E):

$$\Delta \mathbf{w}^{(\infty)} = -\eta\, \nabla E + \mu \left(-\eta\, \nabla E + \mu \left(-\eta\, \nabla E + \mu\, (\dots)\right)\right)$$

$$\Delta \mathbf{w}^{(\infty)} = -\eta\, \nabla E \underbrace{(1 + \mu + \mu^2 + \mu^3 + \cdots)}_{\text{Geometric Series}} = -\left(\frac{\eta}{1-\mu}\right) \nabla E$$

The **momentum parameter μ increases** the effective learning rate

- Remember that $0 \leq \mu < 1$ and therefore $\eta \leq \left(\frac{\eta}{1-\mu}\right)$

How Momentum Helps: Oscillations

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In **directions** that **are steep** (e.g., oscillating sign ∇E):

$$\Delta \mathbf{w}^{(\tau)} = -\eta\, \nabla E + \mu \left(-\eta\, (-\nabla E) + \mu \left(-\eta\, \nabla E + \mu\, (\ldots)\right)\right)$$

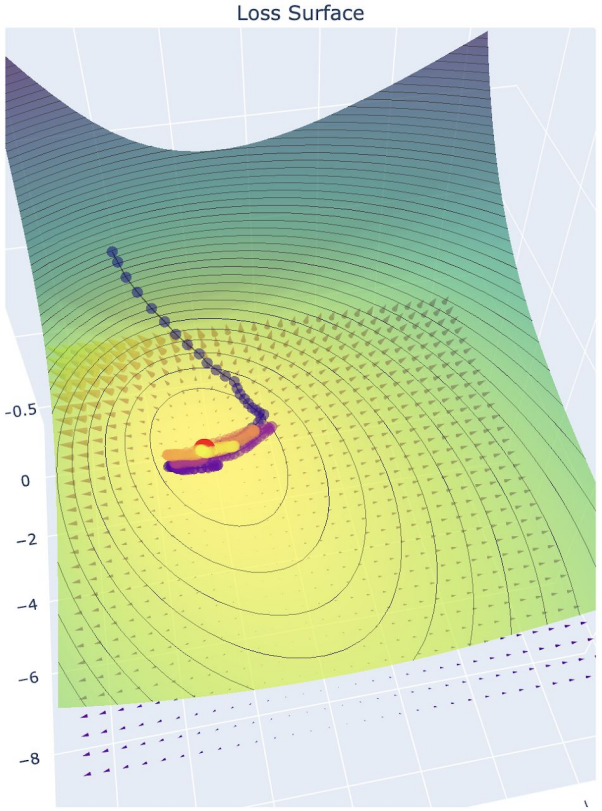
$$\Delta \mathbf{w}^{(\infty)} = -\eta\, \nabla E \underbrace{(1 - \mu + \mu^2 - \mu^3 + \cdots)}_{\text{Alternating Geometric Series}} = -\left(\frac{\eta}{1+\mu}\right) \nabla E$$

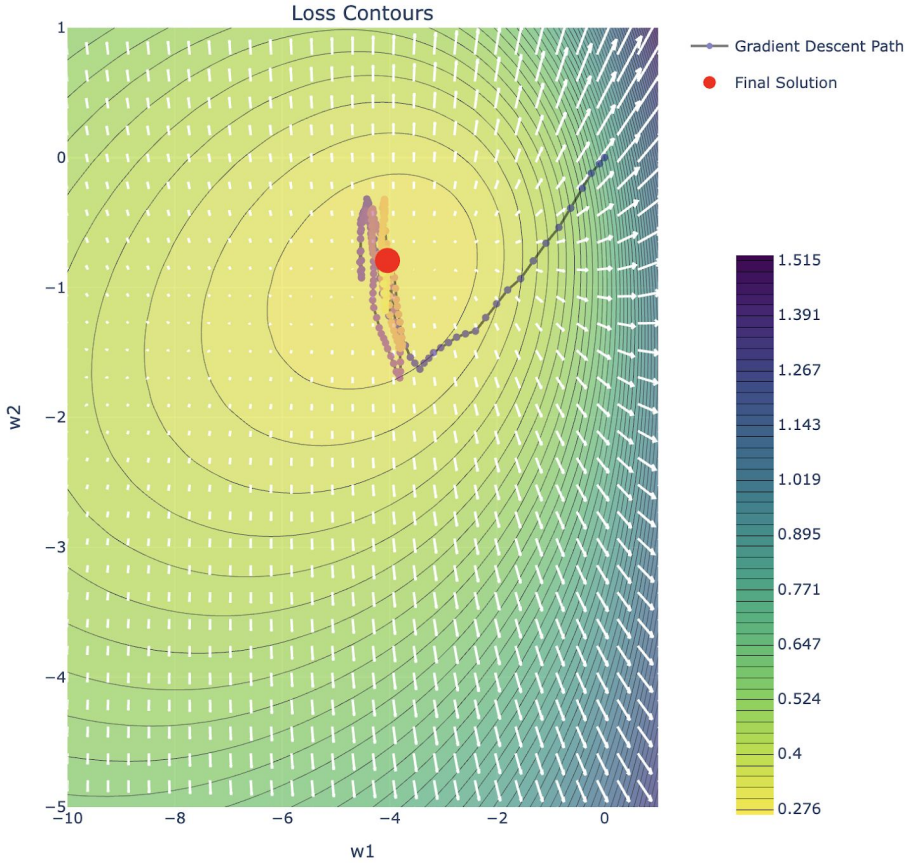
The **momentum parameter μ decrease** the effective learning rate

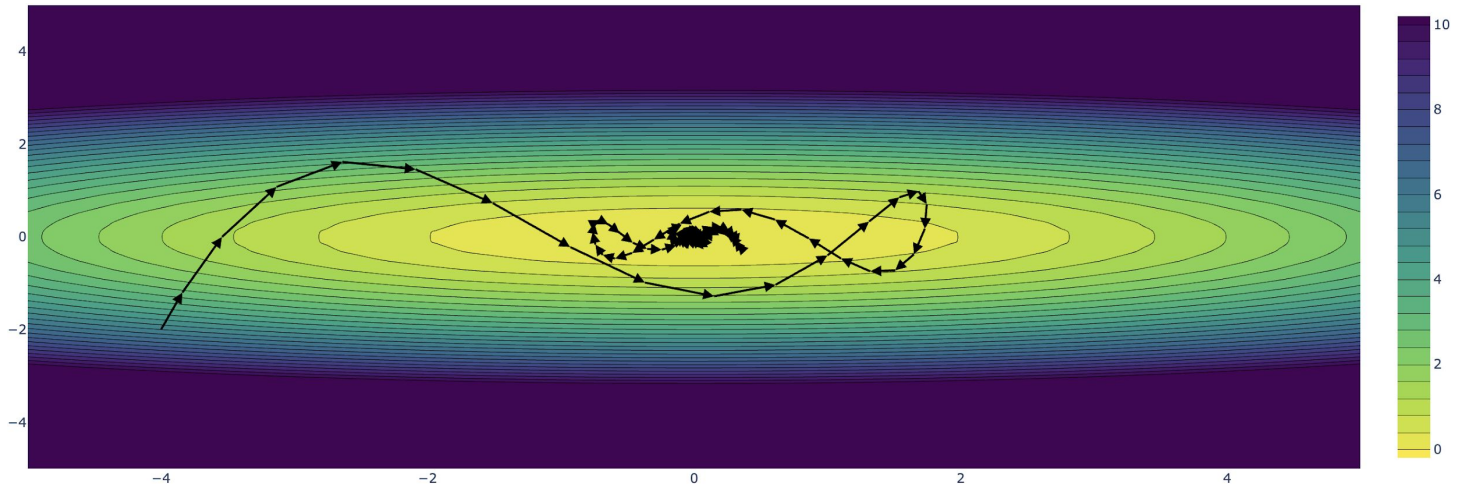
- Remember that $0 \leq \mu < 1$ and therefore $\left(\frac{\eta}{1+\mu}\right) \leq \eta$

Demo

Mini-batch Gradient Descent







Lecture 12

Optimization and Gradient Descent

Credit: Joseph E. Gonzalez and Narges Norouzi

Reference Book Chapters: Chapter 7